

ADVENTURE ARTIST

The Blueprint Book

Every Blueprint in the Unreal project, written from its graphs.

The adventure game, the first-person template,
the prototype actors, and the arena shooter.

Read from the editor dump in Study/_raw 7 October 2026

How to read this book

Adventure Artist contains three layers of Blueprint work. The adventure itself is under AdventureGame: a first-person player, keys, switches, a cube, platforms, fire and spike traps, a level transition, and the screens that end a run. The Artist folder is the art pass of those ideas, plus music, footsteps, a cube spawner, and room logos. Around that work the project still holds Epic's first-person template, a few prototyping actors, the mannequin animation Blueprint, and the Variant Shooter sample.

Each Blueprint is written the way the graph runs. An event or a function is an entry. The calls, branches, casts, loops, and timelines hanging off that entry are the sentences under it. A Get node is folded into the value it feeds, so a wire to a variable becomes the variable's name. Reroute nodes are skipped. When a wire joins a node that was already written, the page points back to that earlier step. When a wire returns to a node still in progress, the page calls it a loop.

Values in parentheses are the pins on that node. A literal is the value stored on the pin. A name is the output of the node on the other end of the wire. Interface calls keep the function name from the node title, such as `fnBPISwitchOn`. Parent calls keep the parent event they invoke.

Animation Blueprints are written differently. A pose graph lists its state machines and players. A transition lists the condition on `bCanEnterTransition`. The event graph of an animation Blueprint is still an execution graph, and it is written that way.

Some records are blank because the dump left them blank. Parent classes, component trees, variable defaults, widget trees, and enum display names are empty for almost every asset. Variables in this book are the ones Get and Set nodes name. `BP_Keyport` did not load. Many control-rig graphs are named and contain no nodes, and those names are listed rather than invented. Nothing here fills a blank the dump left empty.

The adventure comes first. The template and the prototyping actors follow. The arena shooter is the last part, because the adventure character still calls into that sample.

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Part I. The Adventure

Contracts and Shared Functions

These assets are the shared language of the adventure. BPI_Interaction is the on/off call used by switches. BPI_PlayerKeys is the call the player answers when a key is collected or when the HUD asks what is held. Enum_KeyType and Struct_KeyData are the key's type and payload. BPL_FPGame holds the functions other Blueprints call: set a key mesh from the key map, find the player, and play a footstep sound. The stored enum names came back as NewEnumerator0. That token is what the pins contain.

BPI_Interaction

Blueprint /Game/AdventureGame/Designer/Blueprints/Core/BPI_Interaction

fnBPISwitchOn

Declared with no body in this Blueprint.

fnBPISwitchOff

Declared with no body in this Blueprint.

BPI_PlayerKeys

Blueprint /Game/AdventureGame/Designer/Blueprints/Core/BPI_PlayerKeys

fnBPIAddKey

Inputs: Key Type (Enum_KeyType Enum).

Declared with no body in this Blueprint.

fnBPIGetKeys

Returns: Held Keys (Array of Enum_KeyType Enums).

Return Held Keys = NewEnumerator0

Enum_KeyType

/Game/AdventureGame/Designer/Blueprints/Core/Enum_KeyType

This user-defined enum is referenced by the key Blueprints. The dump stored no display names. In the graphs the value is NewEnumerator0.

Struct_KeyData

/Game/AdventureGame/Designer/Blueprints/Core/Struct_KeyData

This user-defined struct is part of the key and weapon data. The dump stored no fields. Graphs that use it pass the struct through as a single value.

BPL_FPGame

Blueprint /Game/AdventureGame/Designer/Blueprints/Core/BPL_FPGame

Variables recovered from the graphs: Start Location, Vector; Max Detection Distance, Float; Player Reference, Pawn; Debug, Boolean; PlayerFound, Boolean; PlayerReference, Pawn; FootStepAsset, BP DA Footsteps.

fnBPLSetKey

Inputs: Static Mesh Array (Array of Static Mesh Component references), KeyMap (Map of Enum_KeyType Enums to Struct Key Data Structures), Key (Enum_KeyType Enum).

Local variables: Player, PlayerFound.

For Each Loop over Static Mesh Array

for each element:

Is Valid (Find (TargetMap = KeyMap, Key = Key).KeyMesh)

if valid:

Set Static Mesh on Array Element (NewMesh = Find (TargetMap = KeyMap, Key = Key).KeyMesh)

Set Material on Array Element (ElementIndex = 0, Material = Find (TargetMap = KeyMap, Key = Key).KeyMaterial)

if not valid:

Set Material on Array Element (ElementIndex = 0, Material = Find (TargetMap = KeyMap, Key = Key).KeyMaterial)

fnBPLFindPlayer

Inputs: Player Reference (Pawn), Start Location (Vector), Max Detection Distance (Float), Debug (Boolean).

Returns: Found (Boolean).

Local variables: Player, PlayerFound.

Is Valid (Player Reference)

if valid:

Sequence

step 1:

Line Trace By Channel (Start = Start Location, End = Get Actor Location on Player Reference, TraceChannel = TraceTypeQuery2, bTraceComplex = false, DrawDebugType = Select(Debug; Option 0 = None, Option 1 = ForDuration), bIgnoreSelf = true)

Branch if ((Break Hit Result (Hit = OutHit).HitActor == Player Reference) AND (Distance (Vector) (V1 = Break Hit Result (Hit = OutHit).TraceStart, V2 = Break Hit Result (Hit = OutHit).TraceEnd) < Max Detection Distance))

if true:

Set PlayerFound = true

if false:

Set PlayerFound = false

step 2:

Return Found = PlayerFound

if not valid:

Return Found = PlayerFound

fnBPLPlayFootStepSound

Inputs: PlayerReference (Pawn).

Local variables: FootStepAsset = DA_Footsteps.

Line Trace By Channel (Start = Get Actor Location on PlayerReference, End = (Get Actor Location on PlayerReference - 0,0,500), TraceChannel = TraceTypeQuery1, bTraceComplex = false, ActorsToIgnore = [PlayerReference],

```
DrawDebugType = None, bIgnoreSelf = true)  
fnGetFootStepSounds on FootStepAsset (PhysicalMaterial = Break Hit Result (Hit = OutHit).PhysMat)  
Play Sound At Location (Sound = SCue, Location = Get Actor Location on PlayerReference, Rotation = 0, 0, 0,  
VolumeMultiplier = 1, PitchMultiplier = 1, StartTime = 0)
```

The Player and the Enemy

BP_AdventureCharacter is the first-person player for the adventure: input, weapons inherited from the shooter sample, keys, damage, and the HUD. BP_Enemy is the opposing character. Both graphs are written here in full, including functions whose names are still the editor defaults NewFunction and Test.

BP_AdventureCharacter

Blueprint /Game/AdventureGame/Designer/Blueprints/Characters/BP_AdventureCharacter

Variables recovered from the graphs: Owned Weapons, Array of BP Weapon Base references; Health, Float; Team Byte, Byte; Player Controller, Player Controller; Current Weapon, BP Weapon Base; Current Bullets, Integer; Mag Size, Integer; Third Person Anim Instance, Anim Instance Class Reference; First Person Anim Instance, Anim Instance Class Reference; Held Keys, Array of Enum_KeyType Enums; HUD, WBP Player HUD; DefaultMovementSpeed, Float; MaxWalkSpeed, Float; SprintMovementSpeed, Float; MID Damage, Material Instance Dynamic; PostProcessSettings, Post Process Settings Structure.

Components recovered from the graphs: FirstPersonMesh, Skeletal Mesh Component; Mesh, Skeletal Mesh Component; CharacterMovement, Character Movement Component; self, Character Movement Component; FirstPersonCamera, Camera Component.

Construction Script

Parent: Construction Script

Event Graph

EnhancedInputAction IA_SwapWeapon.

when triggered:

Branch if (Length (TargetArray = Owned Weapons) > 1)

if true:

Deactivate Weapon on Current Weapon

Set Current Weapon = Owned Weapons[Select Int (A = 0, B = (Find Item (TargetArray = Owned Weapons, ItemToFind = Current Weapon) + 1), bPickA = (Last Index (TargetArray = Owned Weapons) == Find Item (TargetArray = Owned Weapons, ItemToFind = Current Weapon)))]

Activate Weapon on Output_Get

if false:

(stops)

Event Play Firing Montage. Uses Montage.

Montage_Play on Get Anim Instance on FirstPersonMesh (MontageToPlay = Montage, InPlayRate = 1,

ReturnValueType = MontageLength, InTimeToStartMontageAt = 0, bStopAllMontages = true)

Event Add Weapon Class. Uses Weapon Class.

Is Weapon Already Owned (ObjectClass = Weapon Class)

Branch if Owned

if true:

(stops)

if false:

Spawn Actor (Class = Weapon Class, SpawnTransform = Get Actor Transform)

Add (TargetArray = Owned Weapons, NewItem = ReturnValue)

Sequence

step 1:

```

    Is Valid (Current Weapon)
    if valid:
        Deactivate Weapon on Current Weapon
    if not valid:
        (stops)
    step 2:
        Set Current Weapon = ReturnValue
        Activate Weapon on Output_Get
Event Touch Switch Weapon.
Branch if (Length (TargetArray = Owned Weapons) > 1)
    (then the same continuation as the first Branch)
EnhancedInputAction IA_Shoot.
when triggered:
    Get Current Weapon
    then:
        Start Firing on Current Weapon
    else:
        (stops)
when canceled:
    Get Current Weapon
    then:
        Stop Firing on Current Weapon
    else:
        (stops)
when completed:
    Get Current Weapon
    (then the same continuation as the first Get)
Event Touch Shoot Start.
Get Current Weapon
    (then the same continuation as the first Get)
Event fnBPAddKey. Uses Key Type.
Add (TargetArray = Held Keys, NewItem = Key Type)
fnAddKeyHUD on HUD (Key Type = Key Type)
Event Touch Shoot End.
Get Current Weapon
    (then the same continuation as the first Get)
Event Update Weapon HUD. Uses Current Bullets, Magazine Size.
Update Bullet UI on Get HUD on Player Controller (Current Bullets = Current Bullets, Max Bullets = Magazine Size)
Event Add Weapon Recoil. Uses Recoil.
Add Controller Pitch Input (Val = Recoil)
Event Attach Weapon Meshes. Uses Weapon, First Person Mesh, Third Person Mesh.
Attach Actor To Actor on Weapon (ParentActor = self, SocketName = None, LocationRule = SnapToTarget,
RotationRule = SnapToTarget, ScaleRule = SnapToTarget, bWeldSimulatedBodies = TRUE)
Attach Component To Component on First Person Mesh (Parent = FirstPersonMesh, SocketName = HandGrip_R,
LocationRule = SnapToTarget, RotationRule = SnapToTarget, ScaleRule = SnapToTarget, bWeldSimulatedBodies =
TRUE)
Attach Component To Component on Third Person Mesh (Parent = Mesh, SocketName = HandGrip_R, LocationRule =
SnapToTarget, RotationRule = SnapToTarget, ScaleRule = KeepRelative, bWeldSimulatedBodies = TRUE)
Event ReceiveControllerChanged. Uses NewController.
Cast To PlayerController (Object = NewController)
if the cast succeeds:

```

```

Set Player Controller = AsPlayer Controller
Add Mapping Context on Get EnhancedInputLocalPlayerSubsystem (PlayerController = Output_Get)
(MappingContext = IMC_Weapons, Priority = 1, Options =
(bIgnoreAllPressedKeysUntilRelease=True,bForceImmediately=False,bNotifyUserSettings=False))
if the cast fails:
    (stops)
Event Possessed. Uses NewController.
Parent: Possessed (NewController = NewController)
Create Widget (Class = WBP_PlayerHUD)
Set HUD = ReturnValue
fnSetHP on Output_Get (NewHP = Health)
Add To Viewport on HUD (ZOrder = 0)
Event On Weapon Activated. Uses Weapon.
Update Bullet UI on Get HUD on Player Controller (Current Bullets = Current Bullets, Max Bullets = Mag Size)
Set Anim Instance Class on Mesh (NewClass = Third Person Anim Instance)
Set Anim Instance Class on FirstPersonMesh (NewClass = First Person Anim Instance)
Event AnyDamage. Uses Damage.
Set Health = (Health - Damage)
Set Scalar Parameter Value on MID Damage (ParameterName = Damage Amount, Value = Lerp (A = 1, B = 0, Alpha =
(Health / 100)))
Branch if (<= Output_Get)
if true:
    Do Once
    the one time it runs:
        Disable Input (PlayerController = Get Player Controller (PlayerIndex = 0))
        Create Widget (Class = WBP_EliminatedScreen)
        Add To Viewport on ReturnValue (ZOrder = 0)
        Delay (Duration = 5)
        Get Current Level Name (bRemovePrefixString = true)
        Open Level (by Name) (LevelName = ReturnValue, bAbsolute = true)
if false:
    fnSetHP on HUD (NewHP = Health)
    Sequence
    step 1:
        Retriggerable Delay (Duration = 5)
        Branch if (> Health)
        if true:
            Play Damage Blend from the start
            on Update:
                Set Health = ReturnValue
                Set Scalar Parameter Value on MID Damage (ParameterName = Damage Amount, Value = ReturnValue)
                fnSetHP on HUD (NewHP = ReturnValue)
            if false:
                (stops)
        step 2:
            Stop Damage Blend
EnhancedInputAction IA_Sprint.
when triggered:
    Set MaxWalkSpeed = SprintMovementSpeed
when completed:
    Set MaxWalkSpeed = DefaultMovementSpeed

```

Event Destroyed.

Do Once

the one time it runs:

Increment Team Score on Get Game Mode (TeamByte = Team Byte)

Event BeginPlay.

Set DefaultMovementSpeed = MaxWalkSpeed

Create Dynamic Material Instance (Parent = MI_ScreenDamage, OptionalName = None, CreationFlags = None)

Set MID Damage = ReturnValue

Set PostProcessSettings = Make Post Process Settings (WeightedBlendables_Array = [Make Weighted Blendable (Weight = 1, Object = Output_Get)])

Event OnLanded.

fnBPLPlayFootStepSound (PlayerReference = self)

Is Weapon Already Owned

Inputs: ObjectClass (Object Class Reference).

Returns: Owned (Boolean), Found Weapon (BP Weapon Base).

For Each Loop over Owned Weapons

for each element:

Branch if Is A (Object = Array Element, ObjectClass = ObjectClass)

if true:

Return Owned = true, Found Weapon = Array Element

if false:

(stops)

after the loop:

Return Owned = false, Found Weapon = (empty)

NewFunction

Declared with no body in this Blueprint.

Get Weapon Target Location

Returns: Target (Vector).

Line Trace By Channel (Start = Get World Location on FirstPersonCamera, End = (Get World Location on FirstPersonCamera + (Get Forward Vector on FirstPersonCamera * 10000)), TraceChannel = TraceTypeQuery1, bTraceComplex = false, DrawDebugType = None, bIgnoreSelf = true)

Return Target = Select Vector (A = Break Hit Result (Hit = OutHit).ImpactPoint, B = Break Hit Result (Hit = OutHit).TraceEnd, bPickA = Break Hit Result (Hit = OutHit).bBlockingHit)

fnBPGetKeys

Returns: Held Keys (Array of Enum_KeyType Enums).

Return Held Keys = Held Keys

BP_Enemy

Blueprint /Game/AdventureGame/Designer/Blueprints/Characters/BP_Enemy

Variables recovered from the graphs: MaxWalkSpeed, Float; Max Speed, Float; CurrentHP, Float; TotalHP, Float; WaitAtDestination, Float; Base Damage, Float; Eliminated, Boolean; Max Detection Distance, Float; DebugDetection, Boolean; PlayerRef, Character; Found, Boolean.

Components recovered from the graphs: CharacterMovement, Character Movement Component; self, Character Movement Component.

Construction Script

No outgoing execution wire.

Event Graph

Event BeginPlay.

Set MaxWalkSpeed = Max Speed

Set CurrentHP = TotalHP

Set PlayerRef = Get Player Character (PlayerIndex = 0)

Move To Player.

Branch if Found

if true:

AI MoveTo (Pawn = self, Destination = 0, 0, 0, TargetActor = Get Player Character (PlayerIndex = 0),
AcceptanceRadius = 10, bStopOnOverlap = false)

on success:

Delay (Duration = WaitAtDestination)

Move To Player

on fail:

Delay (Duration = 0.2)

Move To Player

if false:

(stops)

On Component Hit (CapsuleComponent). Uses OtherActor.

Branch if (OtherActor == PlayerRef)

if true:

Do Once

the one time it runs:

Apply Damage (DamagedActor = PlayerRef, BaseDamage = Base Damage)

Set Eliminated = true

Delay (Duration = 2)

Destroy Actor

if false:

(stops)

Event Tick.

Branch if NOT Boolean (A = Eliminated)

if true:

fnBPLFindPlayer (Player Reference = Get Player Character (PlayerIndex = 0), Start Location = Get Actor Location,
Max Detection Distance = Max Detection Distance, Debug = DebugDetection)

Set Found = Found

Branch if Output_Get

if true:

Set MaxWalkSpeed = Max Speed

Move To Player

if false:

Set MaxWalkSpeed = 0

if false:

(stops)

Event AnyDamage. Uses Damage.

Set CurrentHP = (CurrentHP - Damage)

Branch if (<= Output_Get)

if true:

Do Once

the one time it runs:

Set Eliminated = true

Delay (Duration = 2)

Destroy Actor

if false:

(stops)

Test

Declared with no body in this Blueprint.

Keys

BP_Key is the pickup. Its construction script asks the function library to dress the mesh from the key map, and its event graph handles what happens when the player overlaps it. BP_Keyport is in the content folder, and the editor dump could not load it.

BP_Key

Blueprint /Game/AdventureGame/Designer/Blueprints/Key/BP_Key

Variables recovered from the graphs: KeyType, Enum_KeyType Enum; KeyMap, Map of Enum_KeyType Enums to Struct Key Data Structures; OtherActor, Actor.

Components recovered from the graphs: KeyMesh, Static Mesh Component.

Construction Script

```
fnBPLSetKey (Static Mesh Array = [KeyMesh], KeyMap = KeyMap, Key = KeyType)
```

Event Graph

On Component Begin Overlap (Capsule). Uses OtherActor.

Set OtherActor = OtherActor

Branch if Does Object Implement Interface (TestObject = Output_Get, Interface = BPI_PlayerKeys)

if true:

 fnBPIAddKey on OtherActor (Key Type = KeyType)

 Delay (Duration = 0.2)

 Destroy Actor

if false:

 (stops)

BP_Keyport

/Game/AdventureGame/Designer/Blueprints/Key/BP_Keyport

The editor dump could not load this Blueprint. The recorded error is: load_asset returned None.

Switches and the Cube

BP_Switch is the overlap switch. The construction script paints it with the off material. The event graph paints the on material and sends fnBPISwitchOn and fnBPISwitchOff to the actors in InteractObjectList. BP_Cube is the cube those systems spawn and move. Its own graphs are included with the rest.

BP_Switch

Blueprint /Game/AdventureGame/Designer/Blueprints/Activation/BP_Switch

Variables recovered from the graphs: OffMaterial, Material Interface; OnMaterial, Material Interface; ActivateOnce, Boolean; InteractObjectList, Array of Actor references.

Components recovered from the graphs: Switch, Static Mesh Component.

Construction Script

Set Material on Switch (ElementIndex = 0, Material = OffMaterial)

Event Graph

On Component Begin Overlap (Trigger).

Set Material on Switch (ElementIndex = 0, Material = OnMaterial)

For Each Loop over InteractObjectList

for each element:

fnBPISwitchOn on Array Element

On Component End Overlap (Trigger).

Branch if ActivateOnce

if true:

(stops)

if false:

Branch if Is Empty (TargetArray = Get Overlapping Actors on Switch (ClassFilter = Actor).OverlappingActors)

if true:

Set Material on Switch (ElementIndex = 0, Material = OffMaterial)

For Each Loop over InteractObjectList

for each element:

fnBPISwitchOff on Array Element

if false:

(stops)

BP_Cube

Blueprint /Game/AdventureGame/Designer/Blueprints/Activation/BP_Cube

Construction Script

No outgoing execution wire.

Event Graph

Unconnected events: Event BeginPlay, Event ActorBeginOverlap, Event Tick.

Platforms

BP_Platform moves its mesh on the timeline TM_MovePlatform between a start location and a target point. fnBPISwitchOn and fnBPISwitchOff start and stop that motion. BP_Platform_Dot is the companion platform Blueprint.

BP_Platform

Blueprint /Game/AdventureGame/Designer/Blueprints/Platforms/BP_Platform

Variables recovered from the graphs: StartLocation, Vector; EndLocation, Vector; TargetPoint, Target Point; Active, Boolean; WaitDuration, Float; TimeToTarget, Float.

Components recovered from the graphs: Platform, Static Mesh Component; TM_MovePlatform, Timeline Component.

Construction Script

No outgoing execution wire.

Event Graph

Event BeginPlay.

Set StartLocation = Get Actor Location

Set EndLocation = Get Actor Location on TargetPoint

Set Play Rate on TM_MovePlatform (NewRate = (1 / TimeToTarget))

Branch if Active

if true:

ev Move Forward

if false:

(stops)

ev Move Forward.

Play TM_MovePlatform

on Update:

Set World Location on Platform (NewLocation = Lerp (Vector) (A = StartLocation, B = EndLocation, Alpha = Alpha), bSweep = false, bTeleport = false)

on Finished:

Delay (Duration = WaitDuration)

Branch if Active

if true:

Branch if (Direction == Forward)

if true:

ev Move Backward

if false:

ev Move Forward

if false:

(stops)

ev Move Backward.

Reverse TM_MovePlatform

ev Stop.

Stop TM_MovePlatform

Event fnBPISwitchOn.
Set Active = true
ev Move Forward
Event fnBPISwitchOff.
Set Active = false
ev Stop

BP_Platform_Dot

Blueprint /Game/AdventureGame/Designer/Blueprints/Platforms/BP_Platform_Dot

Construction Script

Parent: Construction Script

Event Graph

Event BeginPlay.
Parent: BeginPlay
Unconnected events: Event ActorBeginOverlap, Event Tick.

Traps

BP_TrapBase holds the shared damage call, fnApplyDamageToTarget. BP_TrapFire and BP_TrapSpikes are the two traps built on that idea. Where a child only calls the parent, that is the whole graph.

BP_TrapBase

Blueprint /Game/AdventureGame/Designer/Blueprints/Traps/BP_TrapBase

Variables recovered from the graphs: Active, Boolean; TrapTrigger, Box Collision; BaseDamage, Float; DamageInterval, Float; TimerHandler, Timer Handle Structure; OtherActor, Actor.

Construction Script

No outgoing execution wire.

Event Graph

Event BeginPlay.

Branch if Active

if true:

Set Timer by Function Name (FunctionName = fnApplyDamageToTarget, Time = DamageInterval, bLooping = true, bMaxOncePerFrame = false, InitialStartDelay = 0, InitialStartDelayVariance = 0)

Set TimerHandler = ReturnValue

Pause Timer by Handle (Handle = Output_Get)

if false:

(stops)

On Component Begin Overlap (TrapTrigger). Uses OtherActor.

Set OtherActor = OtherActor

fnApplyDamageToTarget

Unpause Timer by Handle (Handle = TimerHandler)

On Component End Overlap (TrapTrigger).

Pause Timer by Handle (Handle = TimerHandler)

fnApplyDamageToTarget

Branch if Active

if true:

For Each Loop over Get Overlapping Actors on TrapTrigger (ClassFilter = Character).OverlappingActors

for each element:

Apply Damage (DamagedActor = Array Element, BaseDamage = BaseDamage)

if false:

(stops)

BP_TrapFire

Blueprint /Game/AdventureGame/Designer/Blueprints/Traps/BP_TrapFire

Variables recovered from the graphs: Active, Boolean; OffMaterial, Material Interface; OnMaterial, Material Interface.

Components recovered from the graphs: TrapBase, Static Mesh Component.

Construction Script

Parent: Construction Script

Event Graph

Event BeginPlay.

Parent: BeginPlay

Event fnBPISwitchOn.

Set Active = false

Set Material on TrapBase (ElementIndex = 0, Material = OffMaterial)

Event fnBPISwitchOff.

Set Active = true

Set Material on TrapBase (ElementIndex = 0, Material = OnMaterial)

Unconnected events: Event ActorBeginOverlap, Event Tick.

BP_TrapSpikes

Blueprint /Game/AdventureGame/Designer/Blueprints/Traps/BP_TrapSpikes

Construction Script

Parent: Construction Script

Event Graph

Event BeginPlay.

Parent: BeginPlay

Unconnected events: Event ActorBeginOverlap, Event Tick.

Leaving the Level

BP_LevelTransition is the Blueprint that changes level. The event graph below is the whole of its logic.

BP_LevelTransition

Blueprint /Game/AdventureGame/Designer/Blueprints/LevelTransition/BP_LevelTransition

Variables recovered from the graphs: OtherActor, Actor; HeldKeys, Array of Enum_KeyType Enums; RequiredKeys, Array of Enum_KeyType Enums; FoundAllKeys, Boolean; NextLevel, World Soft; LevelLoadDelay, Float.

Construction Script

No outgoing execution wire.

Event Graph

On Component Begin Overlap (LevelTrigger). Uses OtherActor.

Set OtherActor = OtherActor

fnBPGetKeys on OtherActor

Set HeldKeys = Held Keys

For Each Loop with Break over RequiredKeys

for each element:

Branch if Contains Item (TargetArray = HeldKeys, ItemToFind = Array Element)

if true:

Set FoundAllKeys = true

if false:

Set FoundAllKeys = false

Break the For Each loop.

after the loop:

Branch if FoundAllKeys

if true:

Sequence

step 1:

Disable Input on OtherActor (PlayerController = Get Player Controller (PlayerIndex = 0))

step 2:

Create Widget (Class = WBP_WinScreen)

Add To Viewport on ReturnValue (ZOrder = 0)

step 3:

Delay (Duration = LevelLoadDelay)

Open Level (by Object Reference) (Level = NextLevel, bAbsolute = true)

if false:

(stops)

Screens

WBP_PlayerHUD is the in-game widget. It exposes fnAddKeyHUD and fnSetHP. WBP_WinScreen and WBP_EliminatedScreen are the end-of-run widgets. The dump did not record their widget trees, so this chapter is the graph logic only.

WBP_PlayerHUD

WidgetBlueprint /Game/AdventureGame/Designer/Blueprints/Widgets/WBP_PlayerHUD

Variables recovered from the graphs: KeysToDisplay, Array of Enum_KeyType Enums; Keys, String; txtKeys, Text Box (Multi- Line); txtHP, Text.

Event Graph

Unconnected events: Event PreConstruct, Event Construct, Event Tick.

fnAddKeyHUD

Inputs: Key Type (Enum_KeyType Enum).

Local variables: Keys.

Add Unique (TargetArray = KeysToDisplay, NewItem = Key Type)

For Each Loop over KeysToDisplay

for each element:

Set Keys = (Keys Append Enum to String (Enumerator = Array Element) Append a new line)

after the loop:

SetText (Multi-Line Text Box) on txtKeys (InText = To Text (String) (InString = Keys))

fnSetHP

Inputs: NewHP (Float).

SetText (Text) on txtHP (InText = To Text (Float) (Value = NewHP))

WBP_WinScreen

WidgetBlueprint /Game/AdventureGame/Designer/Blueprints/Widgets/WBP_WinScreen

Event Graph

Unconnected events: Event PreConstruct, Event Construct, Event Tick.

WBP_EliminatedScreen

WidgetBlueprint /Game/AdventureGame/Designer/Blueprints/Widgets/WBP_EliminatedScreen

Event Graph

Unconnected events: Event PreConstruct, Event Construct, Event Tick.

The Art Track

The Artist folder repeats the gameplay actors for the art pass and adds the pieces that dress a room. BP_Switch_ArtTrack, BP_TrapFire_ArtTrack, and BP_TrapSpikes_ArtTrack sit beside the designer versions. BP_CubeSpawn creates cubes. BP_BGM plays the music. BP_DA_Footsteps chooses a footstep sound. BP_RoomLogo, BP_00, BP_02, and BP_03 are the room marks.

BP_Switch_ArtTrack

Blueprint /Game/AdventureGame/Artist/Blueprints/BP_Switch_ArtTrack

Variables recovered from the graphs: OffMaterial, Material Interface; OnMaterial, Material Interface; ActivateOnce, Boolean; InteractObjectList, Array of Actor references; Trigger, Box Collision.

Components recovered from the graphs: Switch, Static Mesh Component.

Construction Script

Set Material on Switch (ElementIndex = 0, Material = OffMaterial)

Event Graph

On Component Begin Overlap (Trigger).

Set Material on Switch (ElementIndex = 0, Material = OnMaterial)

For Each Loop over InteractObjectList

for each element:

fnBPISwitchOn on Array Element

On Component End Overlap (Trigger).

Branch if ActivateOnce

if true:

(stops)

if false:

Branch if Is Empty (TargetArray = Get Overlapping Actors on Trigger (ClassFilter = Actor).OverlappingActors)

if true:

Set Material on Switch (ElementIndex = 0, Material = OffMaterial)

For Each Loop over InteractObjectList

for each element:

fnBPISwitchOff on Array Element

if false:

(stops)

BP_TrapFire_ArtTrack

Blueprint /Game/AdventureGame/Artist/Blueprints/BP_TrapFire_ArtTrack

Variables recovered from the graphs: FireSpawnRate, Float; Active, Boolean; OffMaterial, Material Interface; OnMaterial, Material Interface.

Components recovered from the graphs: Niagara, Niagara Particle System Component; TrapBase, Static Mesh Component; Audio, Audio Component.

Construction Script

Parent: Construction Script

Set Niagara Variable By String (Float) on Niagara (InVariableName = FireSpawnRate, InValue = FireSpawnRate)

Event Graph

Event fnBPISwitchOn.

Set Active = false

Set Material on TrapBase (ElementIndex = 0, Material = OffMaterial)

Deactivate on Niagara

Stop on Audio

Event fnBPISwitchOff.

Set Active = true

Set Material on TrapBase (ElementIndex = 0, Material = OnMaterial)

Activate on Niagara (bReset = false)

Play on Audio (StartTime = 0)

BP_TrapSpikes_ArtTrack

Blueprint /Game/AdventureGame/Artist/Blueprints/BP_TrapSpikes_ArtTrack

Variables recovered from the graphs: Spike Color, Material Interface.

Components recovered from the graphs: BlockSpike6, Static Mesh Component; BlockSpike5, Static Mesh Component; BlockSpike4, Static Mesh Component; BlockSpike2, Static Mesh Component.

Construction Script

Set Material on BlockSpike6 | BlockSpike5 | BlockSpike4 | BlockSpike2 (ElementIndex = 0, Material = Spike Color)

Parent: Construction Script

Event Graph

Event BeginPlay.

Parent: BeginPlay

Unconnected events: Event ActorBeginOverlap, Event Tick.

BP_CubeSpawn

Blueprint /Game/AdventureGame/Artist/Blueprints/BP_CubeSpawn

Variables recovered from the graphs: SpawnCooldown, Float; CanSpawn, Boolean; CubesSpawned, Integer; MaxNumCubes, Integer; SpawnedCubes, Array of BP Cube references.

Construction Script

No outgoing execution wire.

Event Graph

Event fnBPISwitchOn.

Branch if CanSpawn

if true:

Branch if (CubesSpawned < MaxNumCubes)

if true:

Increment Int on CubesSpawned

```

SpawnActor BP Cube (Class = BP_Cube, SpawnTransform_Location = Get Actor
Transform.ReturnValue_Location, SpawnTransform_Rotation = Get Actor Transform.ReturnValue_Rotation,
SpawnTransform_Scale = 1,1,1)
Add (TargetArray = SpawnedCubes, NewItem = ReturnValue)
Set CanSpawn = false
Ev Reset Spawn
if false:
    Destroy Actor on SpawnedCubes[0]
    Remove Index (TargetArray = SpawnedCubes, IndexToRemove = 0)
    Increment Int on CubesSpawned
    (then the same continuation as the first Increment Int)
if false:
    (stops)
Ev Reset Spawn.
Delay (Duration = SpawnCooldown)
Set CanSpawn = true

```

BP_BGM

Blueprint /Game/AdventureGame/Artist/Blueprints/BP_BGM

Variables recovered from the graphs: Active, Boolean; Volume Multiplier, Float.

Construction Script

No outgoing execution wire.

Event Graph

```

Event BeginPlay.
Branch if Active
if true:
    Play Sound2 D (Sound = MS_BGM, VolumeMultiplier = Volume Multiplier, PitchMultiplier = 1, StartTime = 0,
    bIsUISound = true)
if false:
    (stops)

```

BP_DA_Footsteps

Blueprint /Game/AdventureGame/Artist/Audio/Footsteps/BP_DA_Footsteps

Variables recovered from the graphs: FootStepSounds, Map of Physical Material references to Sound Cue references; DefaultFootSteps, Sound Cue.

fnGetFootStepSounds

Inputs: PhysicalMaterial (Physical Material).

Returns: SCue (Sound Cue).

```

Branch if Find (TargetMap = FootStepSounds, Key = PhysicalMaterial)
if true:
    Return SCue = Find (TargetMap = FootStepSounds, Key = PhysicalMaterial).Value
if false:
    Return SCue = DefaultFootSteps

```

1 graphs are named in this asset and have no nodes in the dump: EventGraph.

BP_RoomLogo

Blueprint /Game/AdventureGame/Artist/Blueprints/Logos/BP_RoomLogo

Construction Script

No outgoing execution wire.

Event Graph

Unconnected events: Event BeginPlay, Event ActorBeginOverlap, Event Tick.

BP_00

Blueprint /Game/AdventureGame/Artist/Blueprints/Logos/BP_00

Construction Script

No outgoing execution wire.

Event Graph

Unconnected events: Event BeginPlay, Event ActorBeginOverlap, Event Tick.

BP_02

Blueprint /Game/AdventureGame/Artist/Blueprints/Logos/BP_02

Construction Script

No outgoing execution wire.

Event Graph

Unconnected events: Event BeginPlay, Event ActorBeginOverlap, Event Tick.

BP_03

Blueprint /Game/AdventureGame/Artist/Blueprints/Logos/BP_03

Construction Script

No outgoing execution wire.

Event Graph

Unconnected events: Event BeginPlay, Event ActorBeginOverlap, Event Tick.

Part II. Templates and Prototypes

The First Person Template

These are Epic's first-person template assets, still in the project beside the adventure. BP_FirstPersonCharacter moves and aims. The controller, camera manager, and game mode are the rest of that pawn. ABP_FP_Copy and CtrlRig_FPWarp are its animation assets. CtrlRig_FPWarp is included with whatever graphs the dump stored.

BP_FirstPersonCharacter

Blueprint /Game/FirstPerson/Blueprints/BP_FirstPersonCharacter

Variables recovered from the graphs: Target Touch UI, User Widget Class Reference.

The character you control in the game, includes firing logic

Construction Script

No outgoing execution wire.

Event Graph

Event Possessed. Uses NewController.

Branch if Is Player Controller on NewController

if true:

Cast To PlayerController (Object = NewController)

if the cast succeeds:

Switch on String (Selection = Get Platform Name)

Default:

Add Mapping Context on Get EnhancedInputLocalPlayerSubsystem (PlayerController = AsPlayer Controller)
(MappingContext = IMC_MouseLook, Priority = 0, Options =
(bIgnoreAllPressedKeysUntilRelease=True,bForceImmediately=False,bNotifyUserSettings=False))

Windows:

Add Mapping Context on Get EnhancedInputLocalPlayerSubsystem (PlayerController = AsPlayer Controller)
(MappingContext = IMC_MouseLook, Priority = 0, Options =
(bIgnoreAllPressedKeysUntilRelease=True,bForceImmediately=False,bNotifyUserSettings=False))

Android:

Create Widget (Class = Target Touch UI, OwningPlayer = AsPlayer Controller)
Add To Player Screen on ReturnValue (ZOrder = 0)

IOS:

Create Widget (Class = Target Touch UI, OwningPlayer = AsPlayer Controller)
(then the same continuation as the first Create Widget)

if the cast fails:

(stops)

if false:

(stops)

EnhancedInputAction IA_MouseLook. Uses ActionValue_X, ActionValue_Y.

when triggered:

Aim (Yaw = ActionValue_X, Pitch = ActionValue_Y)

EnhancedInputAction IA_Look. Uses ActionValue_X, ActionValue_Y.

when triggered:

Aim (Yaw = ActionValue_X, Pitch = ActionValue_Y)

Event Secondary Thumbstick. Uses Axis_X, Axis_Y.

Aim (Yaw = Axis_X, Pitch = Axis_Y)

EnhancedInputAction IA_Move. Uses ActionValue_X, ActionValue_Y.

when triggered:

Move (Left / Right = ActionValue_X, Forward / Backward = ActionValue_Y)

EnhancedInputAction IA_Jump.

when started:

Jump

when completed:

Stop Jumping

Event Primary Thumbstick. Uses Axis_X, Axis_Y.

Move (Left / Right = Axis_X, Forward / Backward = Axis_Y)

Event Touch Jump Start.

Jump

Event Touch Jump End.

Stop Jumping

Move

Inputs: Left / Right (Float), Forward / Backward (Float).

Add Movement Input (WorldDirection = Get Actor Right Vector, ScaleValue = Left / Right, bForce = false)

Add Movement Input (WorldDirection = Get Actor Forward Vector, ScaleValue = Forward / Backward, bForce = false)

Aim

Inputs: Yaw (Float), Pitch (Float).

Add Controller Yaw Input (Val = Yaw)

Add Controller Pitch Input (Val = Pitch)

BP_FirstPersonPlayerController

Blueprint /Game/FirstPerson/Blueprints/BP_FirstPersonPlayerController

Construction Script

No outgoing execution wire.

Event Graph

Event BeginPlay.

Is Valid (Get EnhancedInputLocalPlayerSubsystem)

if valid:

Add Mapping Context on Get EnhancedInputLocalPlayerSubsystem (MappingContext = IMC_Default, Priority = 0,

Options = (bIgnoreAllPressedKeysUntilRelease=True,bForceImmediately=False,bNotifyUserSettings=False))

if not valid:

(stops)

BP_FP_CameraManager

Blueprint /Game/FirstPerson/Blueprints/BP_FP_CameraManager

Construction Script

No outgoing execution wire.

1 graphs are named in this asset and have no nodes in the dump: EventGraph.

GM_FirstPerson

Blueprint /Game/FirstPerson/Blueprints/GM_FirstPerson

GameMode that specified the pawn and HUD Blueprints to use

Construction Script

No outgoing execution wire.

1 graphs are named in this asset and have no nodes in the dump: EventGraph.

ABP_FP_Copy

AnimBlueprint /Game/FirstPerson/Anims/ABP_FP_Copy

Event Graph

Event BlueprintUpdateAnimation has no outgoing execution wire.

Anim Graph

Pose graph: Copy Pose From Mesh; Control Rig.

States and transitions are the graphs that follow.

CtrlRig_FPWarp

ControlRigBlueprint /Game/FirstPerson/Anims/CtrlRig_FPWarp

Rig

The dump has no nodes for this graph.

Touch Controls

BPI_TouchInterface declares the thumbstick and jump messages. UI_Thumbstick is the stick widget, and UI_TouchSimple is the simple touch HUD that binds it.

BPI_TouchInterface

Blueprint /Game/Input/Touch/BPI_TouchInterface

Primary Thumbstick

Inputs: Axis (Vector 2D Structure).

Declared with no body in this Blueprint.

Secondary Thumbstick

Inputs: Axis (Vector 2D Structure).

Declared with no body in this Blueprint.

Touch Jump Start

Declared with no body in this Blueprint.

Touch Jump End

Declared with no body in this Blueprint.

UI_Thumbstick

WidgetBlueprint /Game/Input/Touch/UI_Thumbstick

Variables recovered from the graphs: TrackInput, Boolean; Initial Touch Location_X, Float; Initial Touch Location_Y, Float; Position, Vector 2D Structure; Thumbstick Size_X, Float; Thumbstick Size_Y, Float; Touch Target, ETouchIndex Enum; Invert X, Boolean; Invert Y, Boolean; Scale Output, Float; Container, Size Box; Stick, Image; Thumbstick Size, Vector 2D Structure.

Event Graph

Event PreConstruct.

Set Height Override on Container (InHeightOverride = Thumbstick Size_Y)

Set Width Override on Container (InWidthOverride = Thumbstick Size_X)

Event Tick. Uses MyGeometry.

Set Position = (Local To Viewport (Geometry = MyGeometry, LocalCoordinate = (Thumbstick Size / (X=2,Y=2))).ViewportPosition * Get Viewport Scale)

Branch if TrackInput

if true:

Set TrackInput = Get Input Touch State on Get Player Controller (PlayerIndex = 0) (FingerIndex = Touch Target).bIsCurrentlyPressed

Set Render Translation on Stick (Translation_X = Clamp (Float) (Value = (Get Input Touch State on Get Player Controller (PlayerIndex = 0) (FingerIndex = Touch Target).LocationX - Initial Touch Location_X), Min = ((Thumbstick Size_X / 2) * -1), Max = (Thumbstick Size_X / 2)), Translation_Y = Clamp (Float) (Value = (Get Input Touch State on Get Player Controller (PlayerIndex = 0) (FingerIndex = Touch Target).LocationY - Initial Touch Location_Y), Min = ((Thumbstick Size_Y / 2) * -1), Max = (Thumbstick Size_Y / 2)))

Call Stick Input (NewParam_X = Select Float (A = (Select Float (A = ReturnValue, B = ReturnValue, bPickA = Invert X) * Scale Output), B = 0, bPickA = TrackInput), NewParam_Y = Select Float (A = (Select Float (A = ReturnValue, B = ReturnValue, bPickA = Invert Y) * Scale Output), B = 0, bPickA = TrackInput))

if false:

Set Render Translation on Stick (Translation_X = 0, Translation_Y = 0)

Touch Input Check (FingerIndex = Touch1)

Branch if TrackInput

if true:

(stops)

if false:

Touch Input Check (FingerIndex = Touch2)

Touch Input Check

Inputs: FingerIndex (ETouchIndex Enum).

Set Initial Touch Location_X = Get Input Touch State on Get Player Controller (PlayerIndex = 0) (FingerIndex = FingerIndex).LocationX

Set TrackInput = (Get Input Touch State on Get Player Controller (PlayerIndex = 0) (FingerIndex = FingerIndex).bIsCurrentlyPressed AND (Distance2 D (V1_X = Get Input Touch State on Get Player Controller (PlayerIndex = 0) (FingerIndex = FingerIndex).LocationX, V1_Y = Get Input Touch State on Get Player Controller (PlayerIndex = 0) (FingerIndex = FingerIndex).LocationY, V2 = Position) <= (Thumbstick Size_X Max Thumbstick Size_Y)))

Branch if Output_Get

if true:

Set Touch Target = FingerIndex

if false:

(stops)

Stick Input

Inputs: NewParam (Vector 2D Structure).

Declared with no body in this Blueprint.

UI_TouchSimple

WidgetBlueprint /Game/Input/Touch/UI_TouchSimple

Variables recovered from the graphs: bShowMouseCursor, Boolean; self, Player Controller.

Event Graph

Event Construct.

Bind Event to On Destroyed on Get Controlled Pawn on Get Owning Player (Delegate = OutputDelegate)

Set bShowMouseCursor = true

Set Input Mode Game And UI (PlayerController = Get Owning Player, InMouseLockMode = DoNotLock, bHideCursorDuringCapture = false, bFlushInput = false)

CustomEvent_0.

Remove From Parent

Stick Input (Thumbstick_Move). Uses NewParam.

Primary Thumbstick on Get Controlled Pawn on Get Owning Player (Axis = NewParam)

Stick Input (Thumbstick_Aim). Uses NewParam.

Secondary Thumbstick on Get Controlled Pawn on Get Owning Player (Axis = NewParam)

On Pressed (Btn_Jump).

Touch Jump Start on Get Controlled Pawn on Get Owning Player

On Released (Btn_Jump).

Touch Jump End on Get Controlled Pawn on Get Owning Player

Prototype Actors

The Level Prototyping folder supplies three interactable actors. BP_DoorFrame builds a door from frame meshes and checks keys. BP_JumpPad launches whoever overlaps it. BP_WobbleTarget is the practice target.

BP_DoorFrame

Blueprint /Game/LevelPrototyping/Interactable/Door/BP_DoorFrame

Variables recovered from the graphs: Door Size, Vector; Door Size_X, Float; Door Size_Y, Float; Door Size_Z, Float; Door Frame Scale, Float; Box, Box Collision; Door Detection Adjust, Vector; Door Thickness, Float; Door Frame, Boolean; Split Door, Boolean; Door Frame Override Material, Material Interface; Add Door, Boolean; UseKey, Boolean; KeyMap, Map of Enum_KeyType Enums to Struct Key Data Structures; RequiredKey, Enum_KeyType Enum; Material, Material Interface; NewMesh, Static Mesh; OtherActor, Actor; HasRequiredKey, Boolean; RelativeLocation, Vector; Closed Position, Vector; Closed Position_X, Float; Closed Position_Y, Float; Closed Position_Z, Float; Closed Position_0, Vector; Closed Position_0_X, Float; Closed Position_0_Y, Float; Closed Position_0_Z, Float.

Components recovered from the graphs: Door, Static Mesh Component; Door2, Static Mesh Component; NewMobility, EComponentMobility Enum.

Construction Script

Sequence

step 1:

Branch if Door Frame

if true:

Set Mesh (NewMesh = SM_DoorFrame_Corner, RelativeTransform = Make Transform (Location_X = Break Vector (InVec = Door Size).X, Location_Y = 0, Location_Z = Break Vector (InVec = Door Size).Z, Rotation = 0,180,0, Scale = Make Vector (X = Door Frame Scale, Y = (Break Vector (InVec = Door Size).Y / 100), Z = Door Frame Scale)), Material = Door Frame Override Material, NewMobility = Static)

Set Mesh (NewMesh = SM_DoorFrame_Corner, RelativeTransform = Make Transform (Location_X = (Break Vector (InVec = Door Size).X * -1), Location_Y = 0, Location_Z = Break Vector (InVec = Door Size).Z, Rotation = 0, 0, 0, Scale = Make Vector (X = Door Frame Scale, Y = (Break Vector (InVec = Door Size).Y / 100), Z = Door Frame Scale)), Material = Door Frame Override Material, NewMobility = Static)

Set Mesh (NewMesh = SM_DoorFrame_Edge, RelativeTransform = Make Transform (Location_X = 0, Location_Y = 0, Location_Z = Break Vector (InVec = Door Size).Z, Rotation = 0, 0, 0, Scale_X = ((ReturnValue - ReturnValue) / 100), Scale_Y = (Break Vector (InVec = Door Size).Y / 100), Scale_Z = Door Frame Scale), Material = Door Frame Override Material, NewMobility = Static)

Set Mesh (NewMesh = SM_DoorFrame_Edge, RelativeTransform = Make Transform (Location_X = Door Size_X, Location_Y = 0, Location_Z = (ReturnValue / 2), Rotation = -90,0,0, Scale_X = (ReturnValue / 200), Scale_Y = (Door Size_Y / 100), Scale_Z = Door Frame Scale), Material = Door Frame Override Material, NewMobility = Static)

Set Mesh (NewMesh = SM_DoorFrame_Edge, RelativeTransform = Make Transform (Location_X = ReturnValue, Location_Y = 0, Location_Z = ReturnValue, Rotation = 90,0,0, Scale_X = ReturnValue, Scale_Y = ReturnValue, Scale_Z = Door Frame Scale), Material = Door Frame Override Material, NewMobility = Static)

if false:

(stops)

step 2:

Branch if Add Door

if true:

Set Box Extent on Box (InBoxExtent = (Make Vector (X = Break Vector (InVec = Vector_GetAbs (A = Door Size)).X, Y = Break Vector (InVec = Vector_GetAbs (A = Door Size)).Y, Z = (Break Vector (InVec = Vector_GetAbs (A = Door Size)).Z / 2)) + Door Detection Adjust), bUpdateOverlaps = true)
 Set Relative Location on Box (NewLocation_X = 0, NewLocation_Y = 0, NewLocation_Z = (Break Vector (InVec = Vector_GetAbs (A = Door Size)).Z / 2), bSweep = false, bTeleport = false)

Branch if Split Door

if true:

Add Static Mesh Component [Static Mesh SM_Door] (bManualAttachment = false, RelativeTransform = Make Transform (Location_X = (X / 2), Location_Y = 0, Location_Z = (Z / 2), Rotation = 0, 0, 0, Scale_X = (X / 200), Scale_Y = Door Thickness, Scale_Z = (Z / 200)))

Set Door = ReturnValue

Add Static Mesh Component [Static Mesh SM_Door] (bManualAttachment = false, RelativeTransform = ReturnValue)

Set Door2 = ReturnValue

if false:

Add Static Mesh Component [Static Mesh SM_Door] (bManualAttachment = false, RelativeTransform = Make Transform (Location_X = 0, Location_Y = 0, Location_Z = (Z / 2), Rotation = 0, 0, 0, Scale_X = (X / 100), Scale_Y = Door Thickness, Scale_Z = (Z / 200)))

Set Door = ReturnValue

if false:

Set Collision Enabled on Box (NewType = NoCollision)

step 3:

Branch if UseKey

if true:

fnBPLSetKey (Static Mesh Array = [Door, Door2], KeyMap = KeyMap, Key = RequiredKey)

if false:

(stops)

Event Graph

Event BeginPlay.

Set Closed Position = RelativeLocation

Get Door2

then:

Set Closed Position_0 = RelativeLocation

else:

(stops)

Event ActorBeginOverlap. Uses OtherActor.

Set OtherActor = OtherActor

Branch if (Does Object Implement Interface (TestObject = Output_Get, Interface = BPI_PlayerKeys) AND UseKey)

if true:

Fn Has Key

Branch if KeyFound

if true:

Play Door Control

on Update:

Set Relative Location on Door (NewLocation_X = Select Float (A = Lerp (A = Closed Position_X, B = ReturnValue, Alpha = NewTrack_0), B = 0, bPickA = Split Door), NewLocation_Y = Closed Position_Y, NewLocation_Z = Select Float (A = Closed Position_Z, B = Lerp (A = Closed Position_Z, B = ReturnValue, Alpha = NewTrack_0), bPickA = Split Door), bSweep = false, bTeleport = false)

Get Door2

```

    then:
        Set Relative Location on Door2 (NewLocation_X = ReturnValue, NewLocation_Y = Closed Position_Y,
        NewLocation_Z = Closed Position_Z, bSweep = false, bTeleport = false)
    else:
        (stops)
    if false:
        (stops)
    if false:
        Play Door Control
        Event ActorEndOverlap.
        Branch if (== Length (TargetArray = Get Overlapping Actors on Box (ClassFilter = Character).OverlappingActors))
    if true:
        Reverse Door Control
    if false:
        (stops)

```

Set Mesh

Inputs: NewMesh (Static Mesh), RelativeTransform (Transform), Material (Material Interface), NewMobility (EComponentMobility Enum).

```

Sequence
step 1:
    Add Static Mesh Component [Static Mesh SM_DoorFrame_Corner] (bManualAttachment = false, RelativeTransform
    = RelativeTransform)
    Set Static Mesh on ReturnValue (NewMesh = NewMesh)
    Set Mobility on ReturnValue (NewMobility = NewMobility)
step 2:
    Is Valid (Material)
    if valid:
        Set Material on Add Static Mesh Component [Static Mesh SM_DoorFrame_Corner] (bManualAttachment = false,
        RelativeTransform = RelativeTransform) (ElementIndex = 0, Material = Material)
    if not valid:
        (stops)

```

FnHasKey

Returns: KeyFound (Boolean).

Local variables: HasRequiredKey.

```

Sequence
step 1:
    fnBPGetKeys on OtherActor
    Set HasRequiredKey = Contains Item (TargetArray = Held Keys, ItemToFind = RequiredKey)
step 2:
    Return KeyFound = HasRequiredKey

```

BP_JumpPad

Blueprint /Game/LevelPrototyping/Interactable/JumpPad/BP_JumpPad

Variables recovered from the graphs: Color Target, Linear Color Structure; Velocity, Vector.

Components recovered from the graphs: NS_JumpDemo, Niagara Particle System Component; PointLight, Point Light Component; Cylinder1, Static Mesh Component; self, Movement Component; CharacterMovement, Character Movement Component.

Construction Script

Set Niagara Variable (LinearColor) on NS_JumpDemo (InVariableName = Color, InValue = Color Target)
 Set Light Color on PointLight (NewLightColor = Color Target, bSRGB = true)
 Set Default Custom Primitive Data Float on Cylinder1 (DataIndex = 0, Value = 1)
 Set Default Custom Primitive Data Vector4 on Cylinder1 (DataIndex = 1, Value = Make Vector4 (X = Break Color (InColor = Color Target).R, Y = Break Color (InColor = Color Target).G, Z = Break Color (InColor = Color Target).B, W = Break Color (InColor = Color Target).A))

Event Graph

Event ActorBeginOverlap. Uses OtherActor.
 Cast To Character (Object = OtherActor)
if the cast succeeds:
 Jump on AsCharacter
 Set Velocity = ((Velocity * 1,1,0) + Velocity)
if the cast fails:
 (stops)

BP_WobbleTarget

Blueprint /Game/LevelPrototyping/Interactable/Target/BP_WobbleTarget

Components recovered from the graphs: PhysicsConstraint, Physics Constraint Component; BasePlate, Static Mesh Component; Dummy, Static Mesh Component.

Construction Script

Set Constrained Components on PhysicsConstraint (Component1 = BasePlate, BoneName1 = None, Component2 = Dummy, BoneName2 = None)

Event Graph

Unconnected events: Event ActorBeginOverlap, Event Tick.

The Mannequin

ABP_Unarmed is the mannequin locomotion animation Blueprint: idle, walk and run, jump, fall, and land, plus the event graph that drives those states. The three control rigs are in the project. Where a rig graph has nodes, those nodes are written. Empty rig graphs are named in one list.

ABP_Unarmed

AnimBlueprint /Game/Characters/Mannequins/Anims/Unarmed/ABP_Unarmed

Variables recovered from the graphs: IsFalling, Boolean; GroundSpeed, Float; Strafe, Float; ShouldMove, Boolean; Velocity_X, Float; Velocity_Y, Float; Velocity_Z, Float; Character, Character; Velocity, Vector; Running Speed, Float; bOrientRotationToMovement, Boolean; Can Strafe, Boolean.

Components recovered from the graphs: CharacterMovement, Character Movement Component; MovementComponent, Character Movement Component.

Event Graph

Event BlueprintInitializeAnimation.

Cast To Character (Object = Get Owning Actor)

if the cast succeeds:

Set Character = AsCharacter

Set MovementComponent = CharacterMovement

Set Can Strafe = NOT Boolean (A = bOrientRotationToMovement)

if the cast fails:

(stops)

Event BlueprintUpdateAnimation.

Get Character

then:

Sequence

step 1:

Set Velocity = Velocity

Set GroundSpeed = (Vector Length XY (A = Output_Get) / Running Speed)

Branch if Can Strafe

if true:

Set Strafe = Dot Product (A = Get Actor Right Vector on Character, B = Normalize (A = Velocity, Tolerance = 0))

if false:

(stops)

step 2:

Set ShouldMove = (GroundSpeed > 0.01)

step 3:

Set IsFalling = Is Falling on MovementComponent

else:

(stops)

AnimNotify_AN_FootPlant_Left.

fnBPLPlayFootStepSound (PlayerReference = Get Player Pawn (PlayerIndex = 0))

AnimNotify_AN_FootPlant_Right.

fnBPLPlayFootStepSound (PlayerReference = Get Player Pawn (PlayerIndex = 0))

Anim Graph

Pose graph: state machine Locomotion; state machine Main States; Locomotion; Save cached pose 'Locomotion'; Main States; Slot 'DefaultSlot'; Control Rig.

States and transitions are the graphs that follow.

Idle

This state plays Sequence Player 'MM_Idle'.

Walk / Run

This state plays Blendspace Player 'BS_Idle_Walk_Run'.

Transition

The transition may be taken when ShouldMove.

Transition (2)

The transition may be taken when NOT Boolean (A = ShouldMove).

Locomotion

This state plays Use cached pose 'Locomotion'.

Jump

This state plays Sequence Player 'MM_Jump'.

Transition (3)

The transition may be taken when (IsFalling AND (Velocity_Z > 100)).

Fall Loop

This state plays Sequence Player 'MM_Fall_Loop'.

Transition (4)

The transition may be taken when IsFalling.

Transition (5)

The transition condition has no incoming wire.

Land

This state plays Sequence Player 'MM_Land'.

Transition (6)

The transition may be taken when NOT Boolean (A = IsFalling).

Transition (7)

The transition condition has no incoming wire.

Transition (8)

The transition may be taken when ShouldMove.

2 graphs are named in this asset and have no nodes in the dump: Locomotion, Main States.

CR_Mannequin_Body

ControlRigBlueprint /Game/Characters/Mannequins/Rigs/CR_Mannequin_Body

94 graphs are named in this asset and have no nodes in the dump: Get Negative Transform, Setup Arm, SequenceExecution_SubGraph, Setup Clavicle, Setup Fingers, Setup Leg, SequenceExecution_SubGraph, Setup Foot, SequenceExecution_SubGraph, Forward Arm, SequenceExecution_SubGraph, SequenceExecution_1_SubGraph, Forward Fingers, SequenceAggregate_SubGraph, Forward Clavicle, Forward Leg, SequenceExecution_SubGraph, Forward Foot, SequenceExecution_SubGraph, Set IKFK Visibility, SequenceExecution_SubGraph, Backward Spine, SequenceExecution_SubGraph, Backward Neck, SequenceExecution_SubGraph, Backward Clavicle, Backward Arm, SequenceExecution_SubGraph, Backward Fingers, Backward Leg, SequenceExecution_SubGraph, Backward Foot, SequenceExecution_SubGraph, FK Point At, SequenceExecution_SubGraph, Match 2 Bones IK onto FK, IK 2 Bones, Softness_SubGraph, Exponential Softness_SubGraph, SequenceExecution_SubGraph and 54 more.

CR_Mannequin_FootIK

ControlRigBlueprint /Game/Characters/Mannequins/Rigs/CR_Mannequin_FootIK

FootTrace_SubGraph

The dump has no nodes for this graph.

Rig

The dump has no nodes for this graph.

CR_Mannequin_Procedural

ControlRigBlueprint /Game/Characters/Mannequins/Rigs/CR_Mannequin_Procedural

Rig Graph

The dump has no nodes for this graph.

Part III. The Arena Shooter

Shooter Player, Mode, and HUD

Variant_Shooter is Epic's arena sample, included in this project and partly reused by the adventure character. This chapter is the player pawn, the player controller, the game mode and its score, the shooter interface, and the HUD widgets, including the touch layout.

BP_FPShooter

Blueprint /Game/Variant_Shooter/Blueprints/FirstPerson/BP_FPShooter

Variables recovered from the graphs: Owned Weapons, Array of BP Weapon Base references; Health, Float; Team Byte, Byte; Player Controller, Player Controller; Current Weapon, BP Weapon Base; Current Bullets, Integer; Mag Size, Integer; Third Person Anim Instance, Anim Instance Class Reference; First Person Anim Instance, Anim Instance Class Reference.

Components recovered from the graphs: FirstPersonMesh, Skeletal Mesh Component; Mesh, Skeletal Mesh Component; FirstPersonCamera, Camera Component.

Construction Script

Parent: Construction Script

Event Graph

EnhancedInputAction IA_SwapWeapon.

when triggered:

Branch if (Length (TargetArray = Owned Weapons) > 1)

if true:

Deactivate Weapon on Current Weapon

Set Current Weapon = Owned Weapons[Select Int (A = 0, B = (Find Item (TargetArray = Owned Weapons, ItemToFind = Current Weapon) + 1), bPickA = (Last Index (TargetArray = Owned Weapons) == Find Item (TargetArray = Owned Weapons, ItemToFind = Current Weapon)))]

Activate Weapon on Output_Get

if false:

(stops)

Event Play Firing Montage. Uses Montage.

Montage_Play on Get Anim Instance on FirstPersonMesh (MontageToPlay = Montage, InPlayRate = 1,

ReturnValueType = MontageLength, InTimeToStartMontageAt = 0, bStopAllMontages = true)

Event Add Weapon Class. Uses Weapon Class.

Is Weapon Already Owned (ObjectClass = Weapon Class)

Branch if Owned

if true:

(stops)

if false:

Spawn Actor (Class = Weapon Class, SpawnTransform = Get Actor Transform)

Add (TargetArray = Owned Weapons, NewItem = ReturnValue)

Sequence

step 1:

Is Valid (Current Weapon)

if valid:

```

    Deactivate Weapon on Current Weapon
    if not valid:
        (stops)
    step 2:
        Set Current Weapon = ReturnValue
        Activate Weapon on Output_Get
Event Touch Switch Weapon.
Branch if (Length (TargetArray = Owned Weapons) > 1)
(then the same continuation as the first Branch)
EnhancedInputAction IA_Shoot.
when triggered:
    Get Current Weapon
    then:
        Start Firing on Current Weapon
    else:
        (stops)
when canceled:
    Get Current Weapon
    then:
        Stop Firing on Current Weapon
    else:
        (stops)
when completed:
    Get Current Weapon
    (then the same continuation as the first Get)
Event Touch Shoot Start.
Get Current Weapon
(then the same continuation as the first Get)
Event Touch Shoot End.
Get Current Weapon
(then the same continuation as the first Get)
Event Update Weapon HUD. Uses Current Bullets, Magazine Size.
Update Bullet UI on Get HUD on Player Controller (Current Bullets = Current Bullets, Max Bullets = Magazine Size)
Event Add Weapon Recoil. Uses Recoil.
Add Controller Pitch Input (Val = Recoil)
Event Attach Weapon Meshes. Uses Weapon, First Person Mesh, Third Person Mesh.
Attach Actor To Actor on Weapon (ParentActor = self, SocketName = None, LocationRule = SnapToTarget,
RotationRule = SnapToTarget, ScaleRule = SnapToTarget, bWeldSimulatedBodies = TRUE)
Attach Component To Component on First Person Mesh (Parent = FirstPersonMesh, SocketName = HandGrip_R,
LocationRule = SnapToTarget, RotationRule = SnapToTarget, ScaleRule = SnapToTarget, bWeldSimulatedBodies =
TRUE)
Attach Component To Component on Third Person Mesh (Parent = Mesh, SocketName = HandGrip_R, LocationRule =
SnapToTarget, RotationRule = SnapToTarget, ScaleRule = KeepRelative, bWeldSimulatedBodies = TRUE)
Event ReceiveControllerChanged. Uses NewController.
Cast To PlayerController (Object = NewController)
if the cast succeeds:
    Set Player Controller = AsPlayer Controller
    Add Mapping Context on Get EnhancedInputLocalPlayerSubsystem (PlayerController = Output_Get)
    (MappingContext = IMC_Weapons, Priority = 1, Options =
    (bIgnoreAllPressedKeysUntilRelease=True,bForceImmediately=False,bNotifyUserSettings=False))
if the cast fails:

```

```

    (stops)
Event AnyDamage. Uses Damage.
Set Health = (Health - Damage)
Branch if (<= Output_Get)
if true:
    Sequence
    step 1:
        Is Valid (Current Weapon)
        if valid:
            Deactivate Weapon on Current Weapon
        if not valid:
            (stops)
    step 2:
        Destroy Actor
if false:
    (stops)
Event On Weapon Activated. Uses Weapon.
Update Bullet UI on Get HUD on Player Controller (Current Bullets = Current Bullets, Max Bullets = Mag Size)
Set Anim Instance Class on Mesh (NewClass = Third Person Anim Instance)
Set Anim Instance Class on FirstPersonMesh (NewClass = First Person Anim Instance)
Event Destroyed.
Do Once
the one time it runs:
    Increment Team Score on Get Game Mode (TeamByte = Team Byte)

```

Is Weapon Already Owned

Inputs: ObjectClass (Object Class Reference).

Returns: Owned (Boolean), Found Weapon (BP Weapon Base).

```

For Each Loop over Owned Weapons
for each element:
    Branch if Is A (Object = Array Element, ObjectClass = ObjectClass)
    if true:
        Return Owned = true, Found Weapon = Array Element
    if false:
        (stops)
after the loop:
    Return Owned = false, Found Weapon = (empty)

```

Get Weapon Target Location

Returns: Target (Vector).

```

Line Trace By Channel (Start = Get World Location on FirstPersonCamera, End = (Get World Location on
FirstPersonCamera + (Get Forward Vector on FirstPersonCamera * 10000)), TraceChannel = TraceTypeQuery1,
bTraceComplex = false, DrawDebugType = None, bIgnoreSelf = true)
Return Target = Select Vector (A = Break Hit Result (Hit = OutHit).ImpactPoint, B = Break Hit Result (Hit =
OutHit).TraceEnd, bPickA = Break Hit Result (Hit = OutHit).bBlockingHit)

```

BP_ShooterController

Blueprint /Game/Variant_Shooter/Blueprints/FirstPerson/BP_ShooterController

Construction Script

Parent: Construction Script

Event Graph

Event BeginPlay.

Parent: BeginPlay

Event On Possess. Uses PossessedPawn.

Assign On Destroyed on PossessedPawn (Delegate = OutputDelegate)

OnDestroyed_Event.

Get All Actors Of Class (ActorClass = PlayerStart)

SpawnActor BP FPSooter (Class = BP_FPSooter, SpawnTransform = Get Actor Transform on Random Array Item (TargetArray = OutActors).OutItem)

Possess (InPawn = ReturnValue)

Event Tick has no outgoing execution wire.

GM_Shooter

Blueprint /Game/Variant_Shooter/Blueprints/FirstPerson/GM_Shooter

Variables recovered from the graphs: TeamScores, Map of Bytes to Integers.

Construction Script

No outgoing execution wire.

Event Graph

Event BeginPlay.

Create Widget (Class = UI_Overlay)

Add To Viewport on ReturnValue (ZOrder = 0)

Event IncrementTeamScore. Uses TeamByte.

Add (TargetMap = TeamScores, Key = TeamByte, Value = (Find (TargetMap = TeamScores, Key = TeamByte).Value + 1))

Call Score Updated (NewParam = TeamByte)

Score Updated

Inputs: NewParam (Byte).

Declared with no body in this Blueprint.

GetTeamScore

Inputs: NewParam (Byte).

Returns: NewParam1 (Byte), NewParam2 (Integer).

Return NewParam1 = NewParam, NewParam2 = Find (TargetMap = TeamScores, Key = NewParam).Value

BPI_Shooter

Blueprint /Game/Variant_Shooter/Blueprints/FirstPerson/BPI_Shooter

IncrementTeamScore

Inputs: TeamByte (Byte).

Declared with no body in this Blueprint.

GetTeamScore

Inputs: NewParam (Byte).

Returns: NewParam1 (Byte), NewParam2 (Integer).

Return NewParam1 = 0, NewParam2 = 0

Update Bullet UI

Inputs: Current Bullets (Integer), Max Bullets (Integer).

Declared with no body in this Blueprint.

HUD_Shooter

Blueprint /Game/Variant_Shooter/UI/HUD_Shooter

Variables recovered from the graphs: PlayerOwner, Player Controller; UI_WeaponCounter, UI Weapon Counter.

Construction Script

No outgoing execution wire.

Event Graph

Event BeginPlay.

Create Widget (Class = UI_WeaponCounter, OwningPlayer = PlayerOwner)

Set UI_WeaponCounter = ReturnValue

Add To Player Screen on Output_Get (ZOrder = 0)

Event Update Bullet UI. Uses Current Bullets, Max Bullets.

Update Bullet Counter on UI_WeaponCounter (Mag Size = Max Bullets, Bullet Count = Current Bullets)

Unconnected events: Event ActorBeginOverlap, Event Tick.

GetTeamScore

Inputs: NewParam (Byte).

Returns: NewParam1 (Byte), NewParam2 (Integer).

Return NewParam1 = 0, NewParam2 = 0

UI_Overlay

WidgetBlueprint /Game/Variant_Shooter/UI/UI_Overlay

Variables recovered from the graphs: Team1Score, Text; Team2Score, Text.

Event Graph

Event Construct.

Cast To GM_Shooter (Object = Get Game Mode)

if the cast succeeds:

Bind Event to Score Updated on AsGM Shooter (Delegate = OutputDelegate)

if the cast fails:

(stops)

Get Updated Score. Uses NewParam.

Get Team Score on Get Game Mode (NewParam = NewParam)

SetText (Text) on Select(NewParam1; Option 0 = Team1Score, Option 1 = Team2Score) (InText = To Text (Integer) (Value = NewParam2))

Event PreConstruct has no outgoing execution wire.

UI_WeaponCounter

WidgetBlueprint /Game/Variant_Shooter/UI/UI_WeaponCounter

Variables recovered from the graphs: Image_33, Image; Mag Size, Integer; DM_BulletCount, Material Instance Dynamic; Bullet Count, Integer; Bullet Thickness, Float.

Event Graph

Event PreConstruct.

Create Dynamic Material Instance (Parent = M_BulletCounter, OptionalName = None, CreationFlags = None)

Set DM_BulletCount = ReturnValue

Set Brush From Material on Image_33 (Material = Output_Get)

Update Bullet Counter. Uses Mag Size, Bullet Count.

Set Mag Size = Mag Size

Set Bullet Count = Bullet Count

Set Visibility on Image_33 (InVisibility = Visible)

Set Desired Size Override on Image_33 (DesiredSize_X = (Mag Size * Bullet Thickness), DesiredSize_Y = 40)

Set Scalar Parameter Value on DM_BulletCount (ParameterName = Bullet Max, Value = To Float (Integer) (InInt = Mag Size))

Set Scalar Parameter Value on DM_BulletCount (ParameterName = Bullet Count, Value = To Float (Integer) (InInt = Bullet Count))

BPI_Touch_Shooter

Blueprint /Game/Variant_Shooter/Input/BPI_Touch_Shooter

Touch Shoot Start

Declared with no body in this Blueprint.

Touch Shoot End

Declared with no body in this Blueprint.

Touch Switch Weapon

Declared with no body in this Blueprint.

UI_TouchInterface_Shooter

WidgetBlueprint /Game/Variant_Shooter/Input/UI_TouchInterface_Shooter

Variables recovered from the graphs: bShowMouseCursor, Boolean; self, Player Controller.

Event Graph

Event Construct.

Bind Event to On Destroyed on Get Controlled Pawn on Get Owning Player (Delegate = OutputDelegate)

Set bShowMouseCursor = true

Set Input Mode Game And UI (PlayerController = Get Owning Player, InMouseLockMode = DoNotLock, bHideCursorDuringCapture = false, bFlushInput = false)

Custom Event.

Remove From Parent

Stick Input (Thumbstick_Move). Uses NewParam.

Primary Thumbstick on Get Controlled Pawn on Get Owing Player (Axis = NewParam)
On Pressed (Btn_FireWeapon).
Touch Shoot Start on Get Controlled Pawn on Get Owing Player
Stick Input (Thumbstick_Aim). Uses NewParam.
Secondary Thumbstick on Get Controlled Pawn on Get Owing Player (Axis = NewParam)
On Released (Btn_FireWeapon).
Touch Shoot End on Get Controlled Pawn on Get Owing Player
On Pressed (Btn_SwitchWeapon).
Touch Switch Weapon on Get Controlled Pawn on Get Owing Player
On Pressed (Btn_Jump).
Touch Jump Start on Get Controlled Pawn on Get Owing Player
On Released (Btn_Jump).
Touch Jump End on Get Controlled Pawn on Get Owing Player

Weapons and Pickups

BPI_Pickups and BPI_WeaponHolder are the contracts between the pawn and a weapon. BP_WeaponBase fires the bullet. The pistol, rifle, and grenade launcher are the three children. BP_FirstPersonProjectile and BP_PistolBullet are the projectiles. BP_Pickup spins in the arena until someone collects it. ST_WeaponTable is the weapon data struct. Its fields were not in the dump.

BPI_Pickups

Blueprint /Game/Variant_Shooter/Blueprints/Pickups/BPI_Pickups

Add Weapon

Inputs: Target Weapon (BP Weapon Base Class Reference), Target Context (Input Mapping Context).

Declared with no body in this Blueprint.

BPI_WeaponHolder

Blueprint /Game/Variant_Shooter/Blueprints/Pickups/BPI_WeaponHolder

Attach Weapon Meshes

Inputs: Weapon (BP Weapon Base), First Person Mesh (Skeletal Mesh Component), Third Person Mesh (Skeletal Mesh Component).

Declared with no body in this Blueprint.

Play Firing Montage

Inputs: Montage (Anim Montage).

Declared with no body in this Blueprint.

Add Weapon Recoil

Inputs: Recoil (Float).

Declared with no body in this Blueprint.

Update Weapon HUD

Inputs: Current Bullets (Integer), Magazine Size (Integer).

Declared with no body in this Blueprint.

Get Weapon Target Location

Returns: Target (Vector).

Return Target = 0, 0, 0

Add Weapon Class

Inputs: Weapon Class (BP Weapon Base Class Reference).

Declared with no body in this Blueprint.

On Weapon Activated

Inputs: Weapon (BP Weapon Base).

Declared with no body in this Blueprint.

On Weapon Deactivated

Inputs: *Weapon (BP Weapon Base)*.

Declared with no body in this Blueprint.

On Semi Weapon Refire

Declared with no body in this Blueprint.

BP_WeaponBase

Blueprint /Game/Variant_Shooter/Blueprints/Pickups/BP_WeaponBase

Variables recovered from the graphs: Target Position, Vector; Aim Variance, Float; Current Bullets, Integer; Mag Size, Integer; Firing Montage, Anim Montage; Firing Recoil, Float; Bullet Class, BP First Person Projectile Class Reference; Pawn Owner, Pawn; Time of Last Shot, Float; Is Firing, Boolean; Refire Rate, Float; Refire Timer, Timer Handle Structure; Full Auto, Boolean; Shot Loudness, Float.

Components recovered from the graphs: FP_Weapon, Skeletal Mesh Component; TP_Weapon, Skeletal Mesh Component.

Construction Script

No outgoing execution wire.

Event Graph

Event BeginPlay.

Bind Event to On Destroyed on Get Owner (Delegate = Create Event.OutputDelegate)

Attach Weapon Meshes on Get Owner (Weapon = self, First Person Mesh = FP_Weapon, Third Person Mesh = TP_Weapon)

Cast To Pawn (Object = Get Owner)

if the cast succeeds:

Set Pawn Owner = AsPawn

if the cast fails:

(stops)

On Weapon Owner Destroyed.

Destroy Actor

Start Firing.

Set Is Firing = true

Branch if ((Get Game Time In Seconds - Time of Last Shot) > Refire Rate)

if true:

Fire

if false:

Branch if Full Auto

if true:

Set Timer by Event (Delegate = Create Event.OutputDelegate, Time = (Get Game Time In Seconds - Time of Last Shot), bLooping = false, bMaxOncePerFrame = false, InitialStartDelay = 0, InitialStartDelayVariance = 0)

Set Refire Timer = ReturnValue

if false:

(stops)

Stop Firing.

Set Is Firing = false

Clear and Invalidate Timer by Handle (Handle = Refire Timer)

Fire.

Branch if Is Firing

if true:

Get Weapon Target Location on Get Owner

Fire Bullet (Target Position = Target)

Set Time of Last Shot = Get Game Time In Seconds

Make Noise (Loudness = Shot Loudness, NoiseInstigator = Pawn Owner, NoiseLocation = Get Actor Location on Get Owner, MaxRange = 3000, Tag = Shot)

Branch if Full Auto

if true:

Set Timer by Event (Delegate = Create Event.OutputDelegate, Time = Refire Rate, bLooping = false, bMaxOncePerFrame = false, InitialStartDelay = 0, InitialStartDelayVariance = 0)

Set Refire Timer = ReturnValue

if false:

Set Timer by Event (Delegate = Create Event.OutputDelegate, Time = Refire Rate, bLooping = false, bMaxOncePerFrame = false, InitialStartDelay = 0, InitialStartDelayVariance = 0)

Set Refire Timer = ReturnValue

if false:

(stops)

Fire Cooldown Expired.

On Semi Weapon Refire on Get Owner

Activate Weapon.

Set Actor Hidden In Game (bNewHidden = false)

On Weapon Activated on Get Owner (Weapon = self)

Deactivate Weapon.

Set Actor Hidden In Game (bNewHidden = true)

Stop Firing

On Weapon Deactivated on Get Owner (Weapon = self)

Calculate Bullet Spawn Transform

Inputs: Target Position (Vector).

Returns: NewParam (Transform).

Return NewParam = Make Transform (Location = (Get Socket Location on FP_Weapon (InSocketName = Muzzle) + (Get Forward Vector (InRot = Find Look At Rotation (Start = Get Socket Location on FP_Weapon (InSocketName = Muzzle), Target = Target Position)) * 10)), Rotation = Find Look At Rotation (Start = (Get Socket Location on FP_Weapon (InSocketName = Muzzle) + (Get Forward Vector (InRot = Find Look At Rotation (Start = Get Socket Location on FP_Weapon (InSocketName = Muzzle), Target = Target Position)) * 10)), Target = (Target Position + (Random Unit Vector * Aim Variance))), Scale = 1,1,1)

Fire Bullet

Inputs: Target Position (Vector).

Sequence

step 1:

Spawn Actor (Class = Bullet Class, SpawnTransform = Calculate Bullet Spawn Transform (Target Position = Target Position).NewParam)

step 2:

Play Firing Montage on Get Owner (Montage = Firing Montage)

Add Weapon Recoil on Get Owner (Recoil = Firing Recoil)

step 3:

Sequence

step 1:

```
Decrement Int on Current Bullets
Branch if (<= Result)
if true:
    Set Current Bullets = Mag Size
if false:
    (stops)
step 2:
    Update Weapon HUD on Get Owner (Current Bullets = Current Bullets, Magazine Size = Mag Size)
```

BP_Weapon_Pistol

Blueprint /Game/Variant_Shooter/Blueprints/Pickups/Pistol/BP_Weapon_Pistol

Construction Script

Parent: Construction Script

Event Graph

Event BeginPlay.
Parent: BeginPlay
Event ActorBeginOverlap. Uses OtherActor.
Parent: ActorBeginOverlap (OtherActor = OtherActor)
Event Tick. Uses DeltaSeconds.
Parent: Tick (DeltaSeconds = DeltaSeconds)

BP_Weapon_Rifle

Blueprint /Game/Variant_Shooter/Blueprints/Pickups/Rifle/BP_Weapon_Rifle

Construction Script

Parent: Construction Script

Event Graph

Event BeginPlay.
Parent: BeginPlay
Event ActorBeginOverlap. Uses OtherActor.
Parent: ActorBeginOverlap (OtherActor = OtherActor)
Event Tick. Uses DeltaSeconds.
Parent: Tick (DeltaSeconds = DeltaSeconds)

BP_Weapon_GrenadeLauncher

Blueprint /Game/Variant_Shooter/Blueprints/Pickups/GrenadeLauncher/BP_Weapon_GrenadeLauncher

Construction Script

Parent: Construction Script

Event Graph

Event BeginPlay.
Parent: BeginPlay
Event ActorBeginOverlap. Uses OtherActor.
Parent: ActorBeginOverlap (OtherActor = OtherActor)

Event Tick. Uses DeltaSeconds.

Parent: Tick (DeltaSeconds = DeltaSeconds)

BP_FirstPersonProjectile

Blueprint /Game/Variant_Shooter/Blueprints/Pickups/Projectiles/BP_FirstPersonProjectile

Variables recovered from the graphs: Physics Force, Float; CollisionComponent, Sphere Collision; Instigator, Pawn.

Components recovered from the graphs: Sphere, Static Mesh Component.

Projectile fired by player that pushes physics objects

Construction Script

No outgoing execution wire.

Event Graph

Destroy Projectile.

Delay (Duration = 5)

Destroy Actor

Event Hit. Uses Other, OtherComp, HitLocation.

Sequence

step 1:

Cast To Character (Object = Other)

if the cast succeeds:

Branch if (Other != Get Owner)

if true:

Apply Damage (DamagedActor = Other, BaseDamage = 25, DamageTypeClass = DamageType)

Destroy Projectile

if false:

(stops)

if the cast fails:

Branch if Is Simulating Physics on OtherComp (BoneName = None)

if true:

Add Impulse At Location on OtherComp (Impulse = vector * float (A = Get Velocity, B = Physics Force),

Location = Get Actor Location, BoneName = None)

if false:

(stops)

step 2:

Make Noise (Loudness = 3, NoiseInstigator = Instigator, NoiseLocation = HitLocation, MaxRange = 3000, Tag = None)

Event BeginPlay.

Ignore Actor When Moving on CollisionComponent (Actor = Get Instigator, bShouldIgnore = true)

Timeline Timeline

on Update:

Set Relative Scale3 D on Sphere (NewScale3D = (NewTrack_0 * NewTrack_1 * NewTrack_1 * 0.1))

BP_PistolBullet

Blueprint /Game/Variant_Shooter/Blueprints/Pickups/Projectiles/BP_PistolBullet

Variables recovered from the graphs: CollisionComponent, Sphere Collision.

Components recovered from the graphs: Mobility, EComponentMobility Enum; Sphere, Static Mesh Component.

Construction Script

Parent: Construction Script

Event Graph

Event Hit. Uses MyComp, Other, OtherComp, bSelfMoved, HitLocation, HitNormal, NormalImpulse, Hit.

Sequence

step 1:

Set Collision Enabled on CollisionComponent (NewType = NoCollision)

Switch on EComponentMobility (Selection = Mobility)

Static:

Set Actor Location And Rotation (NewLocation = HitLocation, NewRotation = Make Rot From X (X = (HitNormal * -1)), bSweep = false, bTeleport = false)

Play Wiggle

on Update:

Add Relative Rotation on Sphere (DeltaRotation_Roll = (NewTrack_0 * 3), DeltaRotation_Pitch = 0, DeltaRotation_Yaw = 0, bSweep = false, bTeleport = false)

on Finished:

Delay (Duration = 3)

Set Physics Linear Velocity on Sphere (NewVel = 0, 0, 0, bAddToCurrent = false, BoneName = None)

Set Simulate Physics on Sphere (bSimulate = true)

Stationary:

Set Physics Linear Velocity on Sphere (NewVel = 0, 0, 0, bAddToCurrent = false, BoneName = None)

(then the same continuation as the first Set Physics Linear Velocity)

Movable:

Set Physics Linear Velocity on Sphere (NewVel = 0, 0, 0, bAddToCurrent = false, BoneName = None)

(then the same continuation as the first Set Physics Linear Velocity)

step 2:

Parent: Hit (MyComp = MyComp, Other = Other, OtherComp = OtherComp, bSelfMoved = bSelfMoved, HitLocation = HitLocation, HitNormal = HitNormal, NormalImpulse = NormalImpulse, Hit = Hit)

Event BeginPlay.

Parent: BeginPlay

On Projectile Bounce (ProjectileMovement).

Print String (InString = Hello, bPrintToScreen = true, bPrintToLog = true, TextColor = (R=0,G=0.66,B=1,A=1),

Duration = 2, Key = None)

Unconnected events: Event ActorBeginOverlap, Event Tick.

BP_Pickup

Blueprint /Game/Variant_Shooter/Blueprints/Pickups/BP_Pickup

Variables recovered from the graphs: Spawn Time, Float; SphereCollision, Sphere Collision; Weapon Class, BP Weapon Base Class Reference; Weapon Type, Data Table Row Handle Structure.

Components recovered from the graphs: StaticMesh, Static Mesh Component.

Construction Script

No outgoing execution wire.

Event Graph

Event BeginPlay.

Get Data Table Row (DataTable = Weapon Type.DataTable, RowName = Weapon Type.RowName)

Sequence

step 1:

Async Load Asset (Asset = ReturnValue.StaticMesh)

Completed:

Set Static Mesh on StaticMesh (NewMesh = Cast To StaticMesh (Object = Object).AsStatic Mesh)

step 2:

Async Load Class Asset (AssetClass = ReturnValue.SpawnActor)

Completed:

Cast To BP_WeaponBase Class (Class = Class)

if the cast succeeds:

Set Weapon Class = AsBP Weapon Base

if the cast fails:

(stops)

On Component Begin Overlap (SphereCollision). Uses OtherActor.

Add Weapon Class on OtherActor (Weapon Class = Weapon Class)

Set Hidden In Game on StaticMesh (NewHidden = true, bPropagateToChildren = false)

Set Collision Enabled on SphereCollision (NewType = NoCollision)

Set Actor Tick Enabled (bEnabled = false)

Set Timer by Event (Delegate = OutputDelegate, Time = Spawn Time, bLooping = false, bMaxOncePerFrame = false, InitialStartDelay = 0, InitialStartDelayVariance = 0)

Enable Pickup.

Set Hidden In Game on StaticMesh (NewHidden = false, bPropagateToChildren = false)

Play Spawn In

on Update:

Spin (New Transform Scale = To Vector (Float) (InDouble = ScaleIn))

on Finished:

Set Collision Enabled on SphereCollision (NewType = QueryOnly)

Set Actor Tick Enabled (bEnabled = true)

Event Tick.

Spin (New Transform Scale = 1,1,1)

Spin

Inputs: New Transform Scale (Vector).

Set Relative Transform on StaticMesh (NewTransform_Location_X = 0, NewTransform_Location_Y = 0, NewTransform_Location_Z = (Sin (Radians) (A = (Get Game Time In Seconds * 3)) * 15),

NewTransform_Rotation_Roll = 0, NewTransform_Rotation_Pitch = 0, NewTransform_Rotation_Yaw = (Get Game Time In Seconds * 100), NewTransform_Scale = New Transform Scale, bSweep = false, bTeleport = false)

ST_WeaponTable

/Game/Variant_Shooter/Blueprints/Pickups/ST_WeaponTable

This user-defined struct is part of the key and weapon data. The dump stored no fields. Graphs that use it pass the struct through as a single value.

Shooter AI

BP_FPS shooter_AI is the enemy pawn and BP_AICont is its controller. EnvQueryContext_Target names the query target. The State Tree tasks and the line-of-sight condition are written here with the rest: sensing, facing, shooting, and picking a random float.

BP_FPS shooter_AI

Blueprint /Game/Variant_Shooter/Blueprints/AI/BP_FPS shooter_AI

Variables recovered from the graphs: Weapon, BP Weapon Base; Current HP, Float; CapsuleComponent, Capsule Collision; Current Aim Target, Actor; Is Shooting, Boolean; Weapon Class, BP Weapon Base Class Reference; Team Byte, Byte; Min Aim Offset Z, Float; Max Aim Offset Z, Float; Aim Range, Float; Aim Variance Half Angle, Float; Aim Source, Vector; Aim Target, Vector.

Components recovered from the graphs: CharacterMovement, Character Movement Component; Mesh, Skeletal Mesh Component; FirstPersonMesh, Skeletal Mesh Component; FirstPersonCamera, Camera Component.

Construction Script

Parent: Construction Script

Event Graph

Event BeginPlay.

Parent: BeginPlay

Spawn Actor (Class = Weapon Class, SpawnTransform = Get Actor Transform)

Set Weapon = ReturnValue

Event AnyDamage. Uses Damage.

Branch if (> Current HP)

if true:

Set Current HP = (Current HP - Damage)

Branch if (<= Output_Get)

if true:

Die

if false:

(stops)

if false:

(stops)

Die.

Do Once

the one time it runs:

Call On Death

Increment Team Score on Get Game Mode (TeamByte = Team Byte)

Set Collision Enabled on CapsuleComponent (NewType = NoCollision)

Stop Movement Immediately on CharacterMovement

Stop Active Movement on CharacterMovement

Set Collision Profile Name on Mesh (InCollisionProfileName = Ragdoll, bUpdateOverlaps = true)

Set Simulate Physics on Mesh (bSimulate = true)

Set Physics Blend Weight on Mesh (PhysicsBlendWeight = 1)

Delay (Duration = 5)

Destroy Actor

Event Attach Weapon Meshes. Uses Weapon, First Person Mesh, Third Person Mesh.

Attach Actor To Actor on Weapon (ParentActor = self, SocketName = None, LocationRule = SnapToTarget, RotationRule = SnapToTarget, ScaleRule = SnapToTarget, bWeldSimulatedBodies = TRUE)

Attach Component To Component on First Person Mesh (Parent = FirstPersonMesh, SocketName = HandGrip_R, LocationRule = SnapToTarget, RotationRule = SnapToTarget, ScaleRule = SnapToTarget, bWeldSimulatedBodies = TRUE)

Attach Component To Component on Third Person Mesh (Parent = Mesh, SocketName = HandGrip_R, LocationRule = SnapToTarget, RotationRule = SnapToTarget, ScaleRule = KeepRelative, bWeldSimulatedBodies = TRUE)

Start Shooting. Uses Actor to Shoot.

Set Current Aim Target = Actor to Shoot

Set Is Shooting = true

Start Firing on Weapon

Stop Shooting.

Stop Firing on Weapon

Set Is Shooting = false

Event On Semi Weapon Refire.

Branch if Is Shooting

if true:

Start Firing on Weapon

if false:

(stops)

Has Weapon

Returns: Has (Boolean).

Return Has = Is Valid (Object = Weapon)

On Death

Declared with no body in this Blueprint.

Get Weapon Target Location

Returns: Target (Vector).

Local variables: Aim Source = 0,0,0, Aim Target.

Sequence

step 1:

Set Aim Source = Get World Location on FirstPersonCamera

Is Valid (Current Aim Target)

if valid:

Set Aim Target = (Aim Source + (Random Unit Vector In Cone In Degrees (ConeDir = Normalize (A = ReturnValue, Tolerance = 0), ConeHalfAngleInDegrees = Aim Variance Half Angle) * Aim Range))

if not valid:

Set Aim Target = (Aim Source + (Random Unit Vector In Cone In Degrees (ConeDir = Get Forward Vector on FirstPersonCamera, ConeHalfAngleInDegrees = Aim Variance Half Angle) * Aim Range))

step 2:

Line Trace By Channel (Start = Aim Source, End = Aim Target, TraceChannel = TraceTypeQuery1, bTraceComplex = false, DrawDebugType = None, bIgnoreSelf = true)

Return Target = Select Vector (A = Break Hit Result (Hit = OutHit).ImpactPoint, B = Break Hit Result (Hit = OutHit).TraceEnd, bPickA = Break Hit Result (Hit = OutHit).bBlockingHit)

BP_AICont

Blueprint /Game/Variant_Shooter/Blueprints/AI/BP_AICont

Variables recovered from the graphs: Tags, Array of Names; self, Actor; Team Tag, Name; Current Target Enemy, Actor.

Components recovered from the graphs: StateTreeAI, State Tree AIComponent.

Construction Script

No outgoing execution wire.

Event Graph

Event On Possess. Uses PossessedPawn.

Cast To BP_FPShooter_AI (Object = PossessedPawn)

if the cast succeeds:

Set Tags = [Team Tag]

Bind Event to On Death on AsBP_FPShooter_AI (Delegate = OutputDelegate)

if the cast fails:

(stops)

Set Current Target. Uses Current Target.

Set Current Target Enemy = Current Target

On Death.

Stop Active Movement on Get Path Following Component

Stop Logic on StateTreeAI

Un Possess

Destroy Actor

Clear Current Target.

Set Current Target Enemy = (empty)

Construct Bullet Transform

Inputs: Target Position (Vector), Variance Random (Float).

Returns: NewParam (Transform).

Return NewParam = (empty)

EnvQueryContext_Target

Blueprint /Game/Variant_Shooter/Blueprints/AI/EnvQueryContext_Target

Variables recovered from the graphs: Current Target Enemy, Actor.

ProvideSingleActor

Inputs: QuerierObject (Object Reference), QuerierActor (Actor).

Returns: ResultingActor (Actor).

Cast To BP_AICont (Object = QuerierActor)

if the cast succeeds:

Is Valid (Current Target Enemy)

if valid:

Return ResultingActor = Current Target Enemy

if not valid:

Return ResultingActor = (empty)
if the cast fails:
 Return ResultingActor = (empty)
 1 graphs are named in this asset and have no nodes in the dump: EventGraph.

StateTreeTask_SenseEnemies

Blueprint /Game/Variant_Shooter/Blueprints/AI/StateTreeTask_SenseEnemies

Variables recovered from the graphs: Character, BP FPSShooter AI; Direct Line of Sight Cone Angle, Float; Stimulus Causer, Actor; Target Enemy, Actor; Has Target, Boolean; Investigate Location, Vector; Controller, BP AICont; Has Investigate Location, Boolean; Sense Tag, Name; Last Stimulus Strength, Float.

Components recovered from the graphs: AIPerception, AIPerception Component.

Event Graph

Event EnterState. Uses Transition.

Branch if (Transition.ChangeType == Changed)

if true:

Bind Event to On Target Perception Updated on AIPerception (Delegate = Create Event.OutputDelegate)

Bind Event to On Target Perception Forgotten on AIPerception (Delegate = Create Event.OutputDelegate)

if false:

(stops)

Event ExitState. Uses Transition.

Branch if (Transition.ChangeType == Changed)

if true:

Unbind Event from On Target Perception Updated on AIPerception (Delegate = Create Event.OutputDelegate)

Unbind Event from On Target Perception Forgotten on AIPerception (Delegate = Create Event.OutputDelegate)

if false:

(stops)

On Perception Updated. Uses Actor, Stimulus.

Branch if Actor Has Tag on Actor (Tag = Sense Tag)

if true:

Branch if Stimulus.bSuccessfullySensed

if true:

Has Direct Line of Sight to Stimulus (Stimulus Locaiton = Stimulus.StimulusLocation, Stimulus Causer = Actor)

Branch if Direct Line of Sight

if true:

Set Target Enemy = Actor

Set Current Target on Controller (Current Target = Output_Get)

Set Has Target = true

Set Has Investigate Location = false

if false:

Branch if Is Valid (Object = Target Enemy)

if true:

(stops)

if false:

Branch if (Stimulus.Strength > Last Stimulus Strength)

if true:

Set Last Stimulus Strength = Stimulus.Strength

Set Has Target = false

```

        Set Investigate Location = StimulusLocation
        Set Has Investigate Location = true
    if false:
        (stops)
    if false:
        (stops)
    if false:
        (stops)
    On Perception Forgotten. Uses Actor.
    Branch if (Actor == Target Enemy)
    if true:
        Set Target Enemy = (empty)
        Set Has Target = false
        Set Has Investigate Location = false
        Set Last Stimulus Strength = 0
        Clear Current Target on Controller
        Clear Focus on Controller
    if false:
        Branch if NOT Boolean (A = Is Valid (Object = Target Enemy))
    if true:
        Set Target Enemy = (empty)
        (then the same continuation as the first Set Target Enemy)
    if false:
        (stops)

```

Has Direct Line of Sight to Stimulus

Inputs: Stimulus Locaiton (Vector), Stimulus Causer (Actor).

Returns: Direct Line of Sight (Boolean).

```

    Branch if (Dot Product (A = Normalize (A = (Stimulus Locaiton - Get Actor Location on Character), Tolerance = 0), B =
    Get Actor Forward Vector on Character) >= Cos (Degrees) (A = Direct Line of Sight Cone Angle))
    if true:
        Line Trace By Channel (Start = Get Actor Location on Character, End = Get Actor Location on Stimulus Causer,
        TraceChannel = TraceTypeQuery1, bTraceComplex = false, ActorsToIgnore = [Character, Stimulus Causer],
        DrawDebugType = None, bIgnoreSelf = true)
        Return Direct Line of Sight = NOT Boolean (A = ReturnValue)
    if false:
        Return Direct Line of Sight = false

```

StateTreeCondition_HasLineOfSightToTarget

Blueprint /Game/Variant_Shooter/Blueprints/AI/StateTreeCondition_HasLineOfSightToTarget

Variables recovered from the graphs: Character, BP FPSooter AI; Target, Actor; Line of Sight Cone Angle, Float; Target Half Extent Z, Float; Number of Vertical Line of Sight Checks, Integer; Target Extent Z Offset, Float; Target Center of Mass, Vector; Must Have Line of Sight, Boolean.

Components recovered from the graphs: FirstPersonCamera, Camera Component.

ReceiveTestCondition

Returns: ReturnValue (Boolean).

Local variables: Target Center of Mass, Target Half Extent Z, Target Extent Z Offset.

Sequence*step 1:*

Branch if Is Valid (Object = Target)

if true:

(stops)

if false:

Return ReturnValue = NOT Boolean (A = Must Have Line of Sight)

step 2:

Branch if (Dot Product (A = Get Unit Direction (Vector) (From = Get Actor Location on Character, To = Get Actor Location on Target), B = Get Actor Forward Vector on Character) >= Cos (Degrees) (A = Line of Sight Cone Angle))

if true:

(stops)

if false:

Return ReturnValue = NOT Boolean (A = Must Have Line of Sight)

step 3:

Set Target Center of Mass = Get Actor Bounds on Target (bOnlyCollidingComponents = true, bIncludeFromChildActors = false).Origin

Set Target Half Extent Z = Break Vector (InVec = Get Actor Bounds on Target (bOnlyCollidingComponents = true, bIncludeFromChildActors = false).BoxExtent).Z

Set Target Extent Z Offset = ((Output_Get * 2) / Number of Vertical Line of Sight Checks)

For Loop from 0 to (Number of Vertical Line of Sight Checks - 1)

for each index:

Line Trace By Channel (Start = Get World Location on FirstPersonCamera, End = (Target Center of Mass + 0 + 0 + (Target Half Extent Z - ReturnValue)), TraceChannel = TraceTypeQuery1, bTraceComplex = false,

ActorsToIgnore = [Target], DrawDebugType = None, bIgnoreSelf = true)

Branch if ReturnValue

if true:

(stops)

if false:

Return ReturnValue = Must Have Line of Sight

step 4:

Return ReturnValue = NOT Boolean (A = Must Have Line of Sight)

ReceiveGetDescription

Inputs: Formatting (EStateTreeNodeFormatting Enum).

Returns: ReturnValue (Text).

Return ReturnValue = Select Text (A = HAS Line of Sight to Target, B = Has NO Line of Sight to Target, bPickA = Must Have Line of Sight)

1 graphs are named in this asset and have no nodes in the dump: EventGraph.

StateTreeTask_FaceActor

Blueprint /Game/Variant_Shooter/Blueprints/AI/StateTreeTask_FaceActor

Variables recovered from the graphs: Controller, AIController; Target, Actor.

Event Graph

Event EnterState. Uses Transition.

Branch if (Transition.ChangeType == Changed)

if true:

```

    Set Focus on Controller (NewFocus = Target)
  if false:
    (stops)
  Event ExitState. Uses Transition.
  Branch if (Transition.ChangeType == Changed)
  if true:
    Clear Focus on Controller
  if false:
    (stops)

```

StateTreeTask_FaceLocation

Blueprint /Game/Variant_Shooter/Blueprints/AI/StateTreeTask_FaceLocation

Variables recovered from the graphs: Controller, AIController; Target Location, Vector.

Event Graph

```

Event EnterState. Uses Transition.
Branch if (Transition.ChangeType == Changed)
if true:
  Set Focal Point on Controller (FP = Target Location)
if false:
  (stops)
Event ExitState. Uses Transition.
Branch if (Transition.ChangeType == Changed)
if true:
  Clear Focus on Controller
if false:
  (stops)

```

StateTreeTask_ShootAtTarget

Blueprint /Game/Variant_Shooter/Blueprints/AI/StateTreeTask_ShootAtTarget

Variables recovered from the graphs: Character, BP FPSooter AI; Target, Actor.

Event Graph

```

Event EnterState. Uses Transition.
Branch if (Transition.ChangeType == Changed)
if true:
  Start Shooting on Character (Actor to Shoot = Target)
if false:
  (stops)
Event ExitState. Uses Transition.
Branch if (Transition.ChangeType == Changed)
if true:
  Stop Shooting on Character
if false:
  (stops)

```

ReceiveGetDescription

Inputs: Formatting (EStateTreeNodeFormatting Enum).

Returns: ReturnValue (Text).

Return ReturnValue = Shoot at Target while State is Active

StateTreeTask_SetRandomFloat

Blueprint /Game/Variant_Shooter/Blueprints/AI/StateTreeTask_SetRandomFloat

Variables recovered from the graphs: Min Value, Float; Max Value, Float; Out Value, Float.

Event Graph

Event EnterState. Uses Transition.

Branch if (Transition.ChangeType == Changed)

if true:

Set Out Value = Random Float In Range (Min = Min Value, Max = Max Value)

if false:

(stops)

ReceiveGetDescription

Inputs: Formatting (EStateTreeNodeFormatting Enum).

Returns: ReturnValue (Text).

Return ReturnValue = "Calculate random float between {min} and {max} on Enter" with min = Min Value, max = Max Value

Shooter Animation

These are the first-person and third-person weapon animation Blueprints, the two hand-adjustment control rigs, and the asset-guidelines Blueprint. The hand-adjustment rigs and the guidelines asset are included from the same dump. The animation Blueprints are written as pose graphs, state names, and transition conditions, and their event graphs are written as execution.

ABP_FP_Pistol

AnimBlueprint /Game/Variant_Shooter/Anims/ABP_FP_Pistol

Variables recovered from the graphs: Aim Target, Vector; NewVar_0, Rotator; bIsInAir, Boolean; IsMoving, Boolean; NewVar_X, Float; NewVar_Y, Float; PitchN, Float; NewVar_0_Roll, Float; NewVar_0_Pitch, Float; NewVar_0_Yaw, Float.

Components recovered from the graphs: FirstPersonCamera, Camera Component; First Person Camera, Camera Component; FirstPersonMesh, Skeletal Mesh Component; First Person Mesh, Skeletal Mesh Component.

Event Graph

Event BlueprintBeginPlay.

Cast To BP_FPShooter (Object = Try Get Pawn Owner)

if the cast succeeds:

Set First Person Mesh = FirstPersonMesh

Set First Person Camera = FirstPersonCamera

if the cast fails:

(stops)

Event BlueprintUpdateAnimation.

Is Valid (Try Get Pawn Owner)

if valid:

Sequence

step 1:

Set bIsInAir = Is Falling on Get Movement Component on Try Get Pawn Owner

step 2:

Set IsMoving = float > float (A = Vector Length (A = Get Velocity on Try Get Pawn Owner), B = 0)

step 3:

Line Trace For Objects (Start = Get World Location on First Person Camera, End = (Get World Location on First Person Camera + (Get Forward Vector on First Person Camera * 10000)), ObjectTypes = [ObjectTypeQuery1, ObjectTypeQuery3], bTraceComplex = false, DrawDebugType = None, bIgnoreSelf = true)

Set Aim Target = Select Vector (A = Break Hit Result (Hit = OutHit).Location, B = (Get World Location on First Person Camera + (Get Forward Vector on First Person Camera * 10000)), bPickA = ReturnValue)

step 4:

Set NewVar_X = Dot Product (A = Get Actor Forward Vector on Try Get Pawn Owner, B = Normalize (A = Get Velocity on Try Get Pawn Owner, Tolerance = 0))

step 5:

Set NewVar_0_Roll = 0

Set PitchN = Dot Product (A = Get Forward Vector (InRot = Get Control Rotation on Get Controller on ReturnValue), B = 0,0,1)

if not valid:

(stops)

Anim Graph

Pose graph: Slot 'Arms'; Control Rig; Copy Pose From Mesh.
States and transitions are the graphs that follow.

ABP_FP_Weapon

AnimBlueprint /Game/Variant_Shooter/Anims/ABP_FP_Weapon

Variables recovered from the graphs: Aim Target, Vector; NewVar_0, Rotator; bIsInAir, Boolean; IsMoving, Boolean; NewVar_X, Float; NewVar_Y, Float; PitchN, Float; NewVar_0_Roll, Float; NewVar_0_Pitch, Float; NewVar_0_Yaw, Float.

Components recovered from the graphs: FirstPersonCamera, Camera Component; First Person Camera, Camera Component; FirstPersonMesh, Skeletal Mesh Component; First Person Mesh, Skeletal Mesh Component.

Event Graph

Event BlueprintBeginPlay.

Cast To BP_FPShooter (Object = Try Get Pawn Owner)

if the cast succeeds:

Set First Person Mesh = FirstPersonMesh

Set First Person Camera = FirstPersonCamera

if the cast fails:

(stops)

Event BlueprintUpdateAnimation.

Is Valid (Try Get Pawn Owner)

if valid:

Sequence

step 1:

Set bIsInAir = Is Falling on Get Movement Component on Try Get Pawn Owner

step 2:

Set IsMoving = float > float (A = Vector Length (A = Get Velocity on Try Get Pawn Owner), B = 0)

step 3:

Line Trace For Objects (Start = Get World Location on First Person Camera, End = (Get World Location on First Person Camera + (Get Forward Vector on First Person Camera * 10000)), ObjectTypes = [ObjectTypeQuery1, ObjectTypeQuery3], bTraceComplex = false, DrawDebugType = None, bIgnoreSelf = true)

Set Aim Target = Select Vector (A = Break Hit Result (Hit = OutHit).Location, B = (Get World Location on First Person Camera + (Get Forward Vector on First Person Camera * 10000)), bPickA = ReturnValue)

step 4:

Set NewVar_X = Dot Product (A = Get Actor Forward Vector on Try Get Pawn Owner, B = Normalize (A = Get Velocity on Try Get Pawn Owner, Tolerance = 0))

step 5:

Set NewVar_0_Roll = 0

Set PitchN = Dot Product (A = Get Forward Vector (InRot = Get Control Rotation on Get Controller on ReturnValue), B = 0,0,1)

if not valid:

(stops)

Anim Graph

Pose graph: Slot 'Arms'; Control Rig; Copy Pose From Mesh.
States and transitions are the graphs that follow.

ABP_TP_Pistol

AnimBlueprint /Game/Variant_Shooter/Anims/ABP_TP_Pistol

Variables recovered from the graphs: PitchN, Float; bIsInAir, Boolean; NewVar_X, Float; NewVar_Y, Float; IsMoving, Boolean; NewVar_0_Roll, Float; NewVar_0_Pitch, Float; NewVar_0_Yaw, Float.

Event Graph

Event BlueprintUpdateAnimation.

Is Valid (Try Get Pawn Owner)

if valid:

Sequence

step 1:

Set bIsInAir = Is Falling on Get Movement Component on Try Get Pawn Owner

step 2:

Set IsMoving = float > float (A = Vector Length (A = Get Velocity on Try Get Pawn Owner), B = 0)

step 3:

Set NewVar_X = Dot Product (A = Get Actor Forward Vector on Try Get Pawn Owner, B = Normalize (A = Get Velocity on Try Get Pawn Owner, Tolerance = 0))

step 4:

Is Valid (Get Controller on Try Get Pawn Owner)

if valid:

Set NewVar_0_Roll = 0

Set PitchN = Dot Product (A = Get Forward Vector (InRot = Get Control Rotation on ReturnValue), B = 0,0,1)

if not valid:

(stops)

if not valid:

(stops)

Anim Graph

Pose graph: state machine Locomotio State Machine; Locomotio State Machine; Layered blend per bone; Sequence Player 'MF_Pistol_Idle_ADS'; AimOffset Player 'AO_Pistol'.

States and transitions are the graphs that follow.

FPP_Idle

This state plays Sequence Player 'MM_Idle'.

FPP_JumpStart

This state plays Sequence Player 'MM_Jump'.

FPP_JumpLoop

This state plays Sequence Player 'MM_Fall_Loop'.

AnimationTransitionGraph_213

The transition may be taken when float <= float (A = Get Relevant Anim Time Remaining (FPP_JumpStart), B = 0.1).

FPP_JumpEnd

This state plays Sequence Player 'MM_Land'.

AnimationTransitionGraph_214

The transition may be taken when NOT Boolean (A = bIsInAir).

AnimationTransitionGraph_215

The transition may be taken when float <= float (A = Get Relevant Anim Time Remaining Fraction (FPP_JumpEnd), B = 0.75).

FPP_Run

This state plays Sequence Player 'MF_Unarmed_Jog_Fwd'; Blendspace Player 'BS_Idle_Walk_Run'.

AnimationTransitionGraph_216

The transition may be taken when IsMoving.

AnimationTransitionGraph_217

The transition may be taken when NOT Boolean (A = IsMoving).

AnimationTransitionGraph_218

The transition may be taken when bIsInAir.

AnimationTransitionGraph_219

The transition may be taken when bIsInAir.

1 graphs are named in this asset and have no nodes in the dump: LocomotioStateMachine.

ABP_TP_Rifle

AnimBlueprint /Game/Variant_Shooter/Anims/ABP_TP_Rifle

Variables recovered from the graphs: PitchN, Float; bIsInAir, Boolean; NewVar_X, Float; NewVar_Y, Float; IsMoving, Boolean; NewVar_0_Roll, Float; NewVar_0_Pitch, Float; NewVar_0_Yaw, Float.

Event Graph

Event BlueprintUpdateAnimation.

Is Valid (Try Get Pawn Owner)

if valid:

Sequence

step 1:

Set bIsInAir = Is Falling on Get Movement Component on Try Get Pawn Owner

step 2:

Set IsMoving = float > float (A = Vector Length (A = Get Velocity on Try Get Pawn Owner), B = 0)

step 3:

Set NewVar_X = Dot Product (A = Get Actor Forward Vector on Try Get Pawn Owner, B = Normalize (A = Get Velocity on Try Get Pawn Owner, Tolerance = 0))

step 4:

Is Valid (Get Controller on Try Get Pawn Owner)

if valid:

Set NewVar_0_Roll = 0

Set PitchN = Dot Product (A = Get Forward Vector (InRot = Get Control Rotation on ReturnValue), B = 0,0,1)

if not valid:

(stops)

if not valid:

(stops)

Anim Graph

Pose graph: state machine Locomotio State Machine; Locomotio State Machine; Layered blend per bone; Sequence Player 'MF_Rifle_Idle_ADS'; AimOffset Player 'AO_Rifle'.

States and transitions are the graphs that follow.

FPP_Idle

This state plays Sequence Player 'MM_Idle'.

FPP_JumpStart

This state plays Sequence Player 'MM_Jump'.

FPP_JumpLoop

This state plays Sequence Player 'MM_Fall_Loop'.

AnimationTransitionGraph_213

The transition may be taken when float <= float (A = Get Relevant Anim Time Remaining (FPP_JumpStart), B = 0.1).

FPP_JumpEnd

This state plays Sequence Player 'MM_Land'.

AnimationTransitionGraph_214

The transition may be taken when NOT Boolean (A = bIsInAir).

AnimationTransitionGraph_215

The transition may be taken when float <= float (A = Get Relevant Anim Time Remaining Fraction (FPP_JumpEnd), B = 0.75).

FPP_Run

This state plays Sequence Player 'MF_Unarmed_Jog_Fwd'; Blendspace Player 'BS_Idle_Walk_Run'.

AnimationTransitionGraph_216

The transition may be taken when IsMoving.

AnimationTransitionGraph_217

The transition may be taken when NOT Boolean (A = IsMoving).

AnimationTransitionGraph_218

The transition may be taken when bIsInAir.

AnimationTransitionGraph_219

The transition may be taken when bIsInAir.

1 graphs are named in this asset and have no nodes in the dump: LocomotioStateMachine.

Ctrl_HandAdjusment

ControlRigBlueprint /Game/Variant_Shooter/Anims/Ctrl_HandAdjusment

Rig

The dump has no nodes for this graph.

Ctrl_HandAdjusment_Pistol

ControlRigBlueprint /Game/Variant_Shooter/Anims/Ctrl_HandAdjusment_Pistol

Rig

The dump has no nodes for this graph.

BP_AssetGuidelines_Shooter

Blueprint /Game/Variant_Shooter/Blueprints/BP_AssetGuidelines_Shooter

EventGraph

The dump has no nodes for this graph.

Colophon

This book covers 77 Blueprint assets, 311 graphs, and 2152 nodes. The source is the read-only editor dump in `Study/_raw/assets`, produced by `Study/_tools/dump_blueprints.py`. The book was typeset from that dump on 7 October 2026.